
Repair Manual

911 Carrera 4 **(993)**

Volume II: Transmission Manual

II Manual Transmission (911 Carrera 4)

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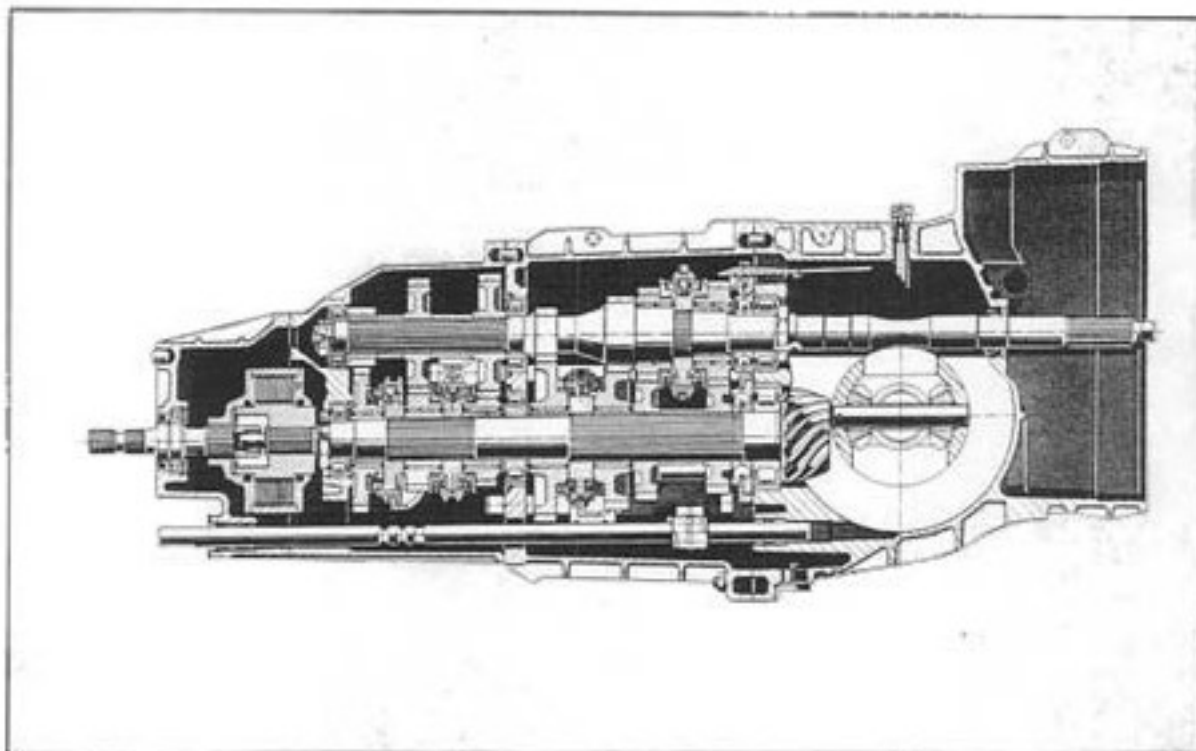
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3 Technical data

6 speed manual transmission G 64

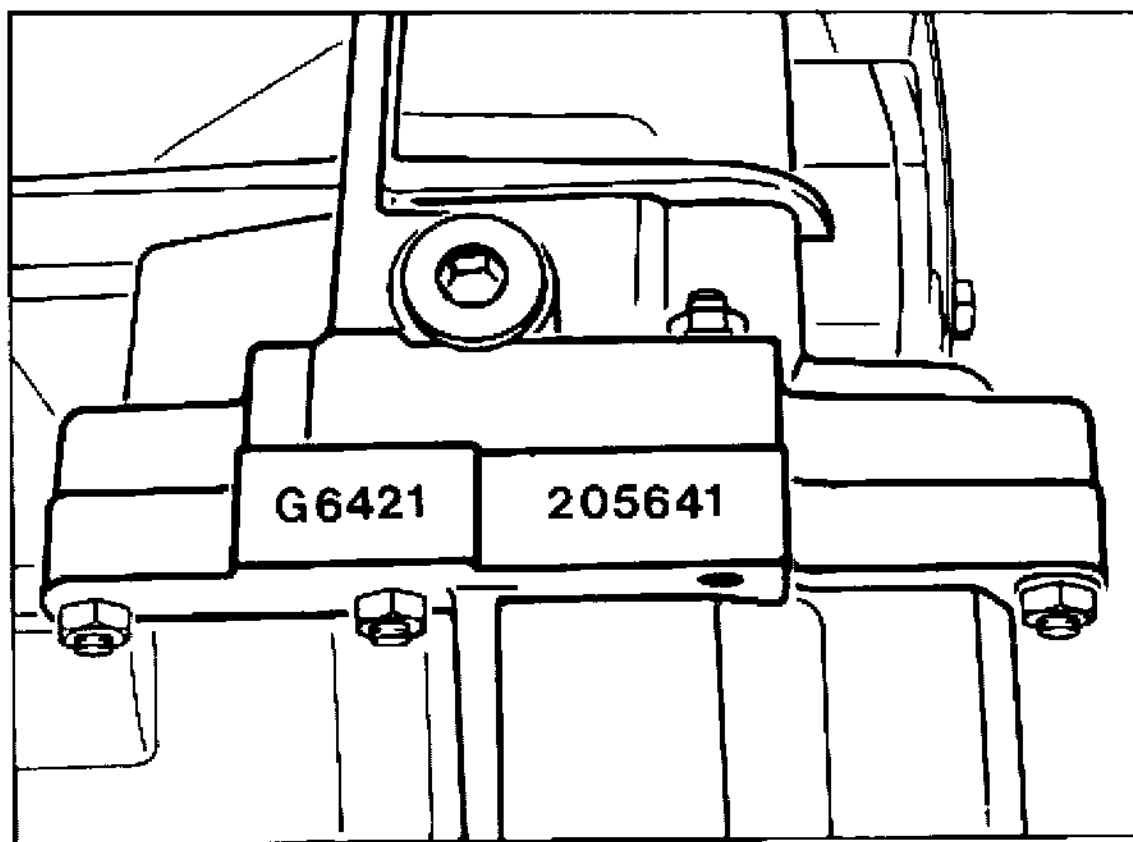
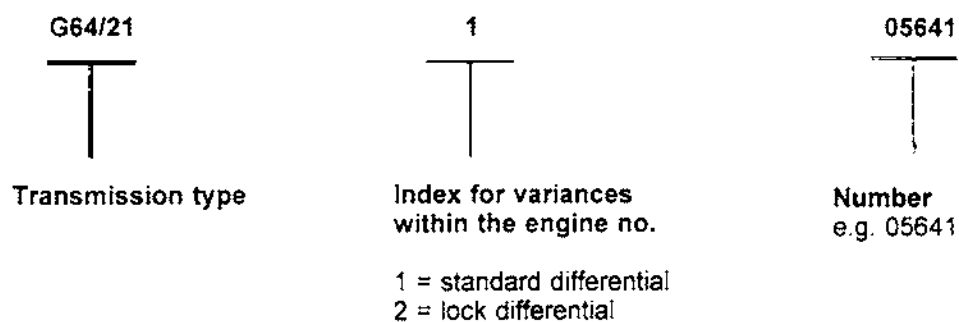


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Type	Equipment	installed in:	Model year
G 64/21	6 speed	911 Carrera 4 worldwide except USA, CH, A	'95 / '96
G 64/20	6 speed	911 Carrera 4 USA, CH, A	'95 / '96
		911 Carrera 4 worldwide	'97

3 Technical Data

Structure of transmission numbers



3 Technical Data (Manual transmission)

General Data		Manual transmission G 64 / 21 / 20	
Transmission ratios	G64/21 $Z_1 : Z_2 = Z_2 : Z_1$	G64/20 $Z_1 : Z_2 = Z_2 : Z_1$	
1st gear	11 : 42 = 3.818	11 : 42 = 3.818	
2nd gear	20 : 43 = 2.150	21 : 43 = 2.048	
3rd gear	25 : 39 = 1.560	27 : 38 = 1.407	
4th gear	33 : 41 = 1.242	34 : 38 = 1.118	
5th gear	41 : 42 = 1.024	42 : 39 = 0.928	
6th gear	39 : 32 = 0.820	40 : 31 = 0.775	
Reverse	14 : 40 = 2.857	14 : 40 = 2.857	
Final drive	Hypoid bevel-gear drive with 16 mm offset		
Final drive ratio	9 : 31 = 3.444	9 : 31 = 3.444	
Capacity	3.8 liters multigrade transmission oil SAE 75 W 90 of API classification GL 5 (or MIL-L 2105 B)		

3 Technical Data

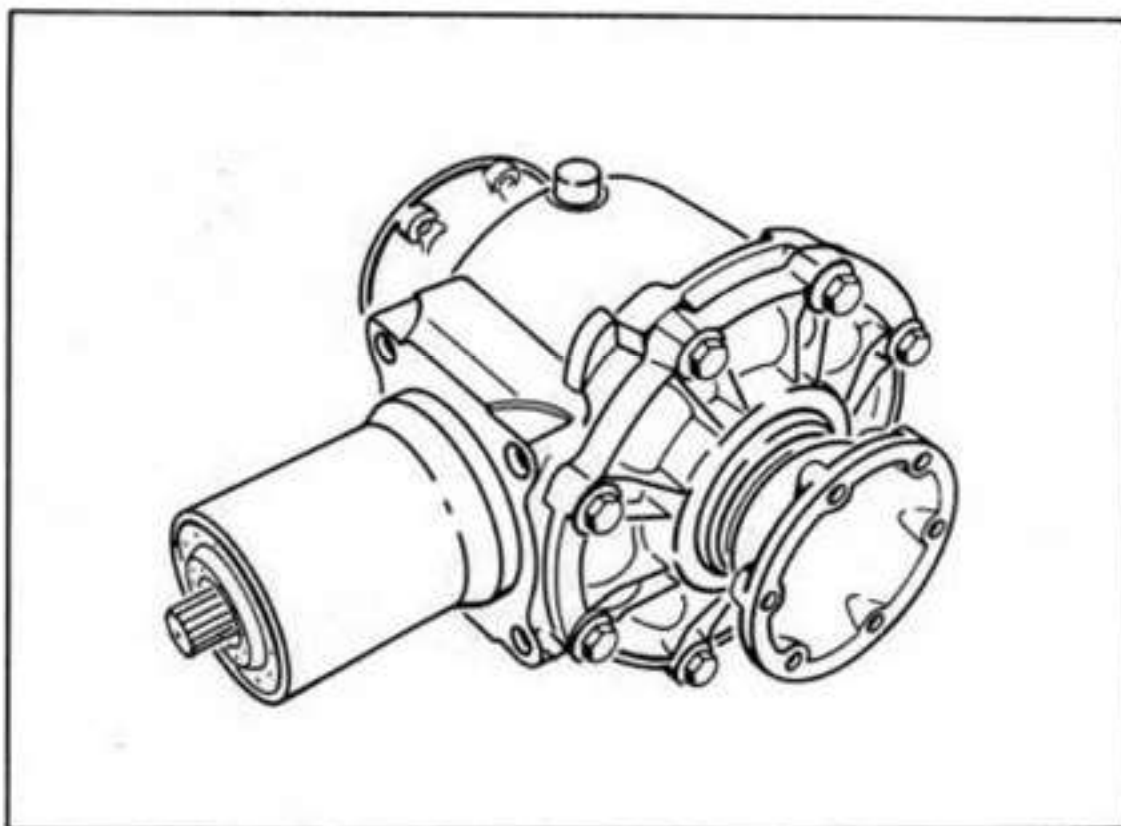
Torque specifications for manual transmission, front axle final drive, transmission suspension and central tube

Location	Thread	Tightening torque Nm (ftlb)
Oil drain and oil filler plugs	M 22 x 1,5	30
Hexagon head nuts at: Front and side transmission cover, gear and transmission housings. Tensioning plate	M 8	23
Clamping plate to intermediate case	M 6	10
Hexagon head nut / input shaft	M 22 x 1,5	200
Hexagon head nut / input shaft	M 30 x 1,5	250
Hexagon head nut / output shaft	M 30 x 1,25	300
Hexagon head nut / shift fork	M 8	23
Reverse light switch to gear housing	M 18 x 1,5	35
Guide tube to transmission housing	M 6	10
Selector gate to tensioning plate	M 6	10
Collar nut / return gear II	M 8	23
Vent to transmission housing	M 14 x 1,5	35

Location	Thread	Tightening torque Nm (ftlb)
Hexagon head bolt / joint flange	M 10	44 (32)
Oil pan to tension plate	M 5 (microencapsulated)	6 (4)
Ring gear to differential housing (hexagon head bolt with ribbed seating surface)	M 12 x 1,25	200 (147)
Front axle final drive		
Oil filler plug	M 22	22 (16)
Oil drain plug	M 22	30 (22)
Final drive mount at rear axle		
Crossmember to body	M 10 x 70	46 (34)
Mount to crossmember (lock nut)	M 10	46 (34)
Mount to transmission (lock nut)	M 10	46 (34)
Final drive mount at front axle		
Front axle final drive mount to front axle crossmember	M 8	23 (17)
Front axle final drive mount to central tube	M 12	85 (63)
Central tube		
Double clamp to central shaft	M 8	35...40 (26...29)
Manual transmission to central tube	M 12	85 (63)
Front axle final drive to central tube	M 10	46 (34)

3 Technical Data

Front axle final drive

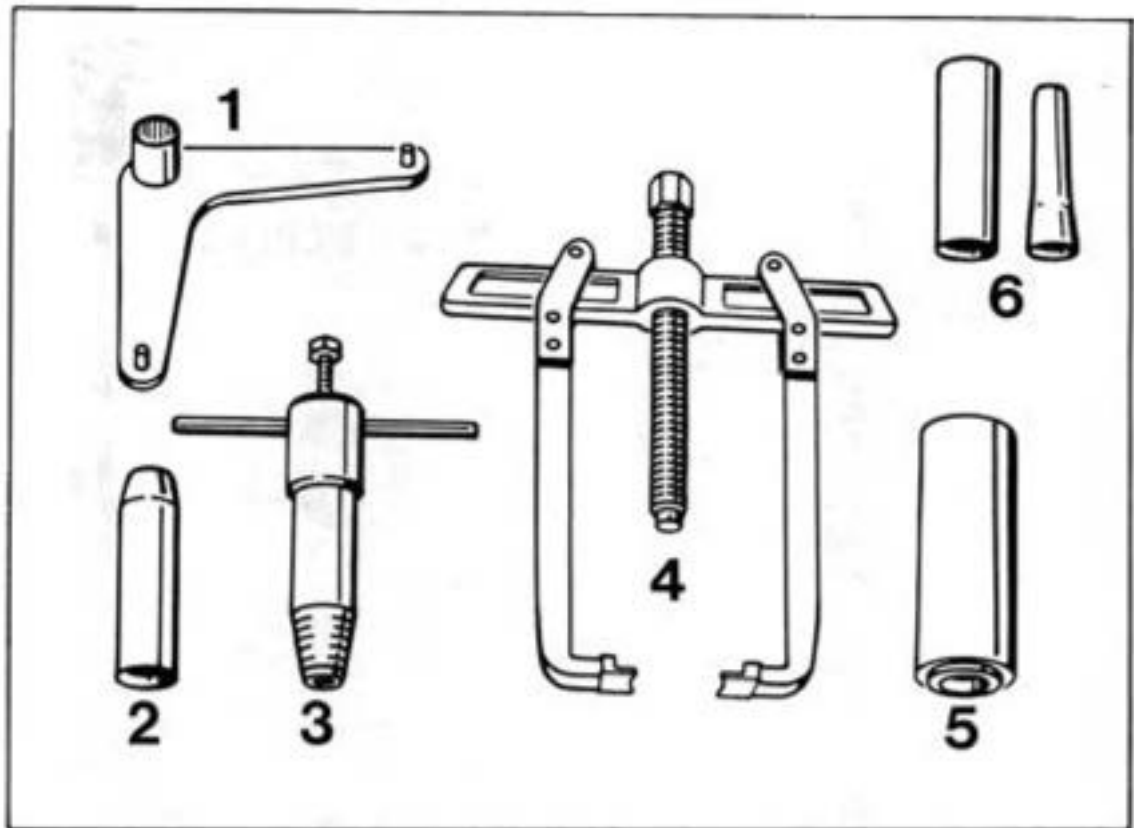


Type	Code letter	Equipment	Installed in:	Model Year
Z64/20			911 Carrera 4 worldwide	'95

General data	Front axle final drive Z64/20
Final drive ratio ($Z_2 : Z_1 = i$)	31 : 9 = 3.444
Final drive	Bevel drive without hypoid offset
Capacity	Approx. 0.6 liters hypoid oil 75 W 90 of API classification GL 5 (or MIL-L 2105 B)

34 35 37 Dismantling and assembling transmission

Tools

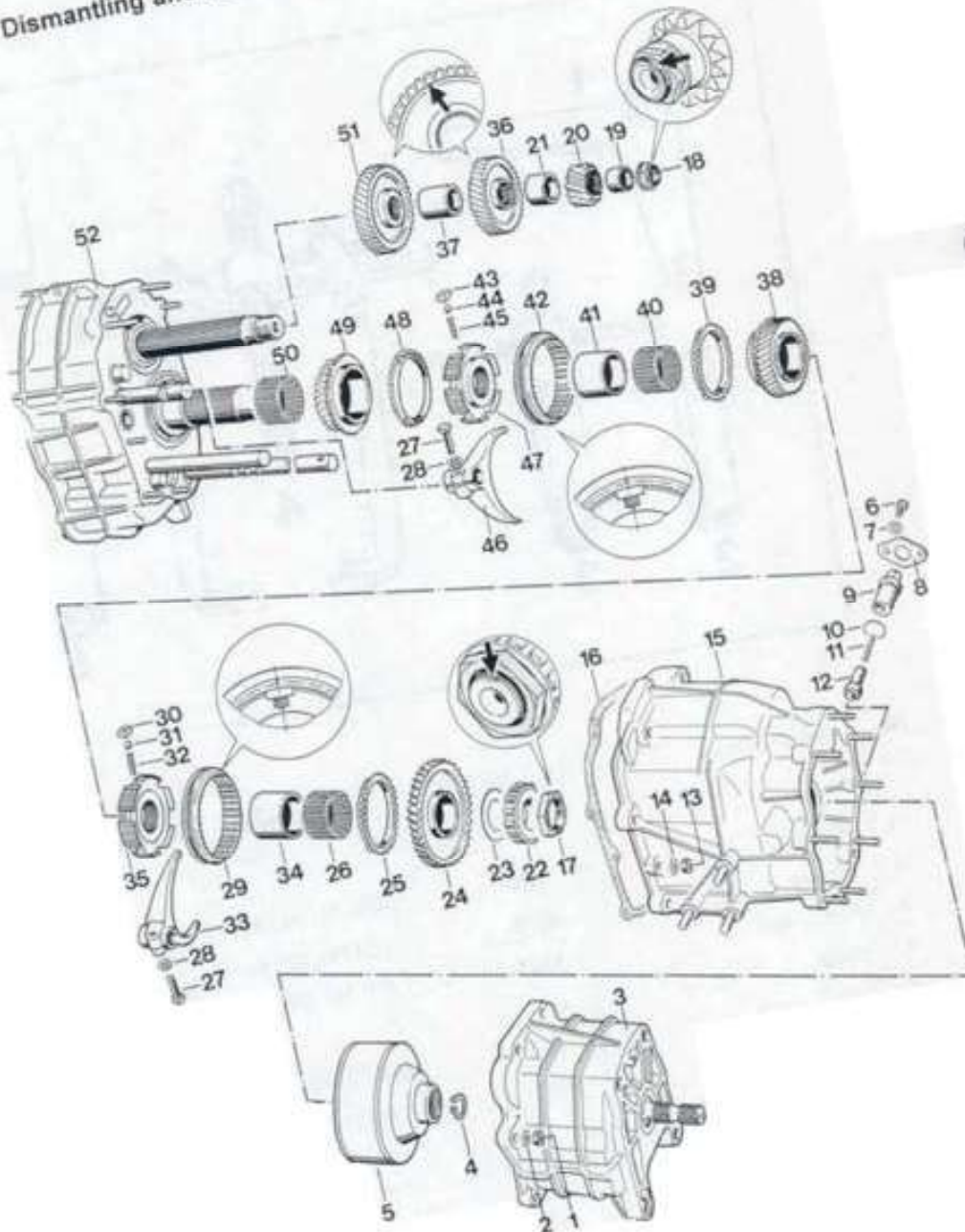


2009-34

No.	Designation	Special tool	Order number	Explanation
1	Bracket	9253	000.721.925.30	with arms of puller 9284
2	Sleeve	9255	000.721.925.50	
3	Pulling tool	9251	000.721.925.10	
4	Puller	-	-	
5	Thrust piece	9256	000.721.925.60	
6	Assembly tool	9547	000.721.954.70	

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Dismantling and assembling transmission



No.	Designation	Qty.	Removal	Note:
				Installation
1	Hexagon head nut	8		MA = 23 Nm (17 ftlb.)
2	Washer	8		
3	Transfer casing	1		Seal between transfer casing and intermediate casing is made with Loctite 573
4	Snap ring	1	Remove with suitable pointed pliers	Install with Special Tool 9547
5	Viscous clutch	1		
6	Hexagon head bolt	2		Tighten to 10 Nm (7 ftlb.)
7	Washer	2		
8	Tensioning plate	1		
9	Eccentric bushing	1		Adjust
10	Seal	1		Replace
11	Thrust spring	1		
12	Locking bushing	1		
13	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
14	Washer	10		
15	Intermediate casing	1		
16	Gasket	1		Replace
17	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 300 Nm (221 ftlb.). Upset flange to lock
18	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 200 Nm (147 ftlb.). Upset flange to lock
19	Bearing inner race	1	Pull off across fixed gear No. 20	Heat to approx. 120 °C
20	Fixed gear (Reverse)	1		
21	Spacer sleeve	1		
22	Cylindrical roller bearing	1	Pull off across loose gear No. 24	Heat to approx. 120 °C
23	Thrust washer	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
24	Loose gear (Reverse)	1		
25	Synchronizer ring	1	Mark for reinstallation	Check for wear. Fit with the same gear-wheel (Cogs must face driver dogs)
26	Needle bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
27	Hexagon head bolt	2		Tighten to 23 Nm (17 ftlb.)
28	Washer	2		
29	Selector sleeve	1	Lift off along with shift fork No. 33. Take care not to lose synchromesh parts.	Center the centerpunch marks relative to the driver dogs. Offset side faces loose gear No. 24
30	Driver dog	3		Domed end faces selector sleeve
31	Ball	3		
32	Spring	3		
33	Shift fork (Reverse)	1		Adjust so that play is barely felt at the selector sleeve when reverse is engaged. When neutral is engaged, the synchronizing ring must be able to rotate freely
34	Inner race**	1	Mark for reinstallation. Pull off across guide sleeve No. 35	Heat to approx. 120 °C
35	Guide sleeve (with snap ring)	1		Open end of snap ring must not be located in the driver dog area. Install in correct position complete with selector sleeve and shift fork (Snap ring faces loose gearwheel No. 38)

No.	Designation	Qty.	Note:	
			Removal	Installation
36	Fixed gear (5th gear)	1		Identification groove faces fixed gear No. 20. Do not confuse with fixed gear No. 51
37	Spacer sleeve	1		
38	Loose gear (5th gear)	1		Do not confuse with loose gear No. 49
39	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. (Cogs face driver dogs)
40	Needle roller bearing cage	1	Mark for reinstallation	Fit with the same gearwheel
41	Inner race**	1	Mark for reinstallation. Pull off with arms of Special Tool 9284 across loose gear No. 49	Fit with the same gearwheel. Heat to approx. 120 °C
42	Selector sleeve	1		Center the centerpunch marks relative to the driver dogs
43	Driver dog	3		Domed side faces selector sleeve
44	Ball	3		
45	Spring	3		
46	Shift fork (5th and 6th gears)	1		Adjust. In neutral position, selector sleeve must be set exactly in the middle between the loose gearwheels
47	Guide sleeve	1		Install complete with selector sleeve and shift fork
48	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gear (Cogs face driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
49	Loose gear (6th gear)	1		Do not confuse with loose gear No. 38
50	Needle roller bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
51	Fixed gear (6th gear)	1		Identification groove faces fixed gear No. 36. Do not confuse with fixed gear No. 36
52	Transmission	1		

* Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

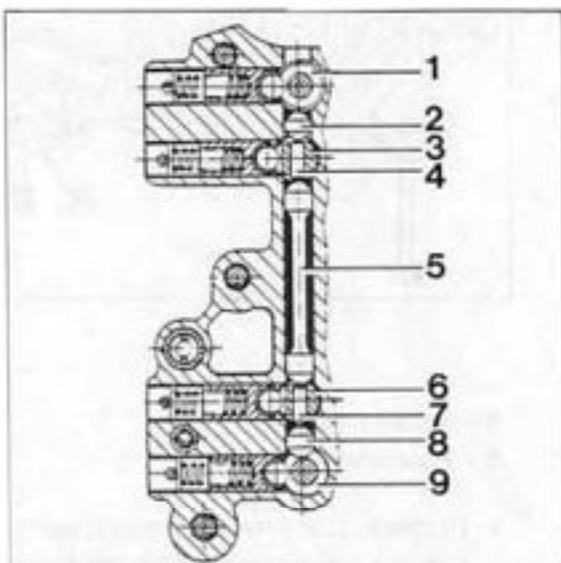
Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

** A result of tolerances, the bearing inner rings (no. 34 and 41) may either from a force fit on the shaft or have a certain amount of play.

Dismantling and assembly notes

Note

If the shift rods are moved across the neutral or gear latching positions when the transmission is dismantled or assembled, the small intermediate locks may drop out inadvertently.

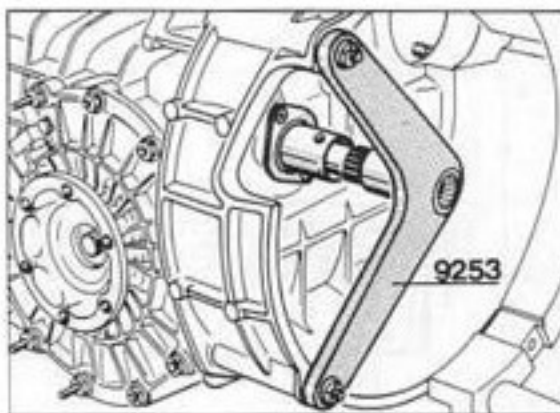


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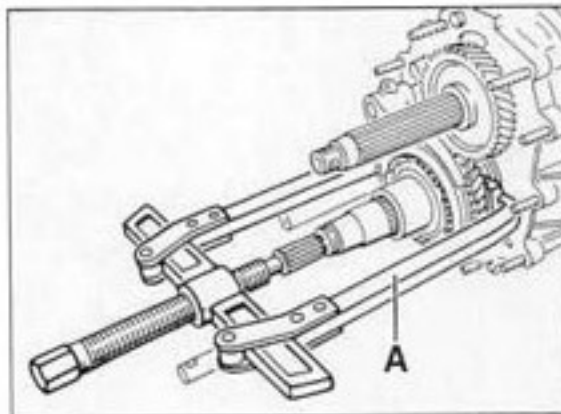
4 and 7 = Intermediate locks

Dismantling

1. Engage 6th gear, lock drive shaft with Special Tool **9253** and unscrew hexagon head nuts for output and input shafts.



2. Engage 4th gear and pull off inner race of needle roller bearing (5th gear).



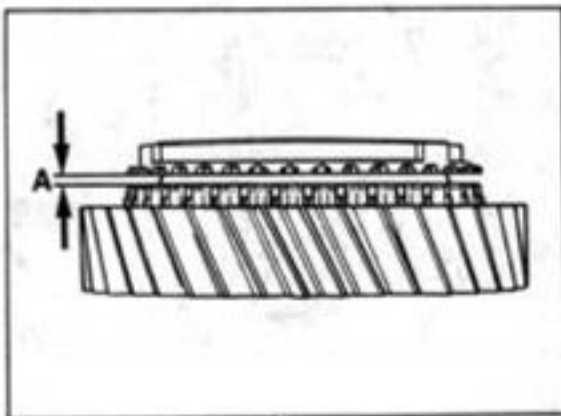
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A = Arms of puller No. 9284

Assembly

1. Check synchronizing rings by pushing rings onto tapers of gearwheels and measuring gap „A“ with a feeler gauge.

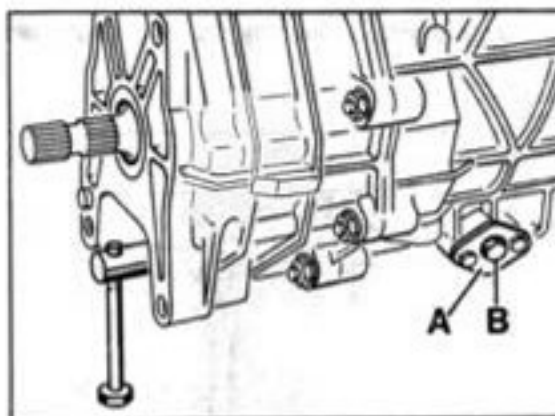
Installation dimension (new) = 0.9 mm min.
Wear limit = 0.6...0.7 mm



518-35

2. Engage 4th gear and fit all gearwheels.

3. Adjust inner shift rod until all gears may be preselected freely without notching. To adjust, undo hexagon head bolts of tensioning plate and rotate eccentric bushing.

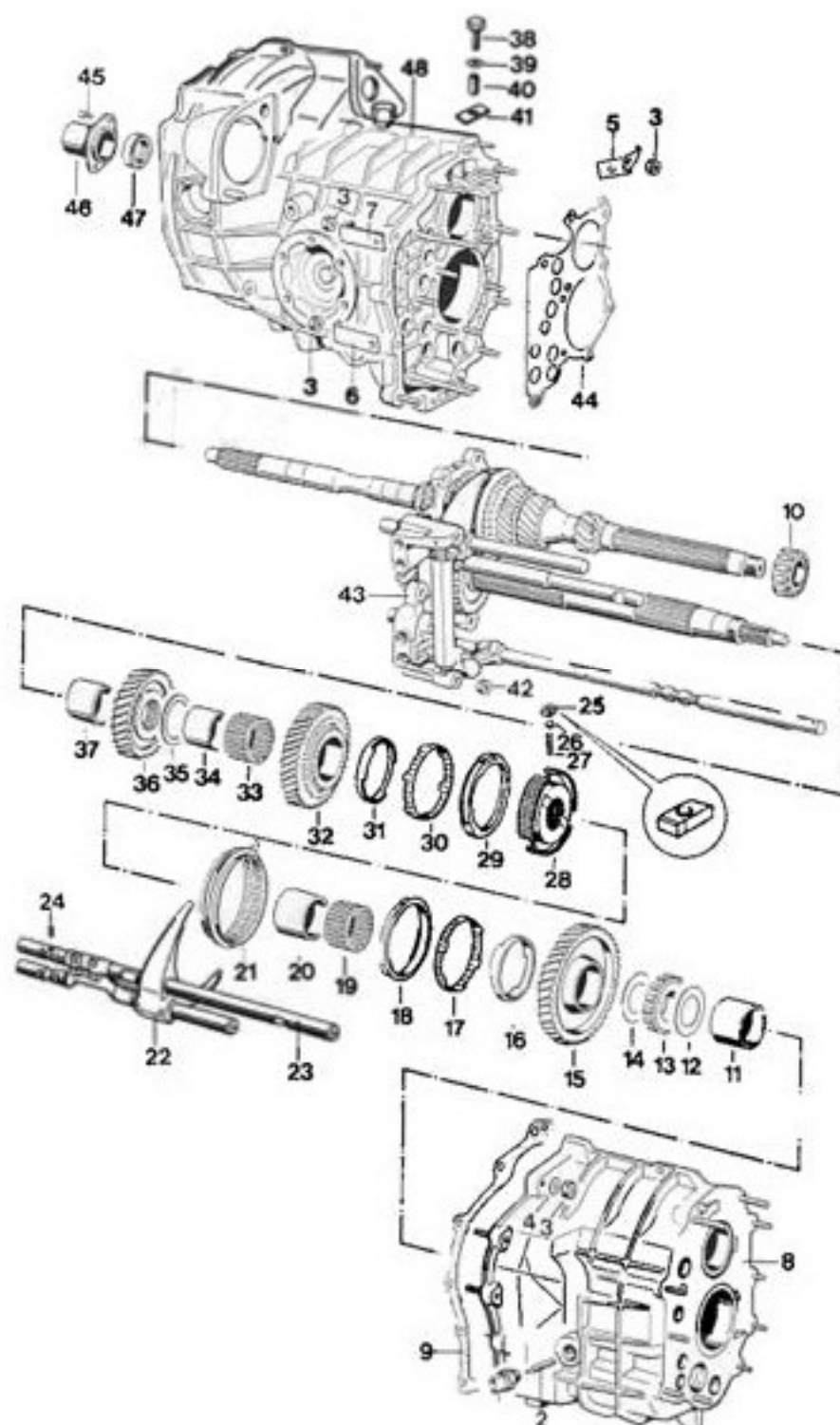


2014-34

A = Tensioning plate
B = Eccentric bushing

4. To check, shift through all gears, using a long M 8 bolt that is screwed into the inner shift rod.

34 35 37 Dismantling and assembling transmission



No.	Designation	Qty.	Removal	Note:	
				Installation	
1	Reverse light switch	1		Tighten to 35 Nm (26 ftlb.)	
2	Plunger	1		Stepped end faces switch	
3	Hexagon head nut	12		Tighten to 23 Nm (17 ftlb.)	
4	Washer	10			
5	Retaining bracket	1	deleted		
6	Retaining bracket (straight)	1			
7	Retaining bracket (domed)	1			
8	Gear housing	1			
9	Gasket	1		Replace	
10	Cylindrical roller bearing	1	Pull off	Heat to approx. 120° C	
11	Inner race	1	Mark for reinstallation. Pull off across loose gear-wheel No. 15. Observe clearance at reverse shift rod	Fit with the same gear-wheel. Heat to approx. 120°C	
12	Thrust washer	1			
13	Cylindrical roller bearing	1		Heat to approx. 120° C	
14	Thrust washer	1			
15	Loose gearwheel (1st gear)	1			
16	Friction ring	1	Mark for reinstallation	Fit with the same gear-wheel	
17	Tapered ring	1	Mark for reinstallation	Fit with the same gear-wheel. Tabs must engage in the cutouts in the loose gearwheel	

No.	Designation	Qty.	Note:	
			Removal	Installation
18	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tapered ring. Three lugs face driver dogs
19	Needle roller bearing	1	Mark for reinstallation	Fit with the same gear-wheel
20	Inner race	1	Shift rods in neutral position. Pull out shift rod for reverse gear. Pull off across loose gearwheel No. 32. Mark for reinstallation.	Fit with the same gear-wheel. Heat to approx. 120 °C
21	Shift sleeve	1		Insert complete with guide sleeve and shift rods. The missing tooth of the internal teeth of the guide sleeve (No. 28) must be aligned exactly above the oil bore of the output shaft. For further notes, refer to page 35 - 213
22	Shift rod with pinned shift fork	1		
23	Shift rod (reverse)	1		
24	Spacer	1		Apply stiff grease to insert into shift rod
25	Driver dog	3		Place into correct position
26	Ball	3		
27	Spring	3		

No.	Designation	Qty.	Note:	
			Removal	Installation
28	Guide sleeve	1		Missing tooth of internal teeth must be placed exactly above oil bore of output shaft
29	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered ring. The lugs face the driver dogs.
30	Tapered ring	1	Mark for reinstallation	Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
31	Friction ring	1	Mark for reinstallation	Fit with the same gear-wheel
32	Loose gearwheel (2nd gear)	1		
33	Needle roller bearing	1	Mark for reinstallation	Fit with the same gear-wheel
34	Inner race	1	Mark for reinstallation. Pull off across fixed gear-wheel No. 36	Fit with the same gear-wheel. Heat to approx. 120 °C
35	Thrust washer	1		
36	Fixed gearwheel (3rd gear)	1		Large flange faces thrust washer No. 35
37	Spacer sleeve	1		
38	Hexagon head bolt (micro-sealed)	1		Must always be replaced. Tightening torque 6 Nm (4 ftlb.)
39	Washer	1		
40	Spacer sleeve	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
41	Retaining plate	1		Tighten to 23 Nm (17 ftlb.)
42	Hexagon head nut	10		
43	Tensioning plate with gear set	1		
44	Adjusting washer "S3"	X		Redetermine thickness if required
45	Oval-head screw	2	Note number and thickness for reinstallation	Tighten to 10 Nm (7 ftlb.)
46	Guide tube	1		
47	Shaft seal	1		
48	Transmission housing	1		Do not fit until gear set has been fitted (refer to page 35 - 1)

Note

Due to the tolerances (transition fit), the inner bearing rings (nos. 11, 20 and 34) may have clearance or press-fit on the shaft.

Dismantling and assembly notes

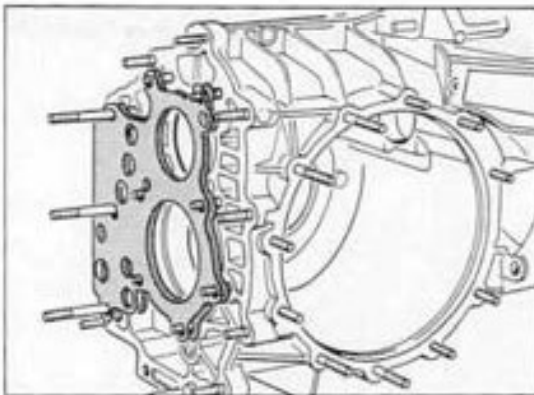
Dismantling

Note

To facilitate assembly, do not remove the complete gear set assembly but partially dismantle the installed drive pinion.

Assembly

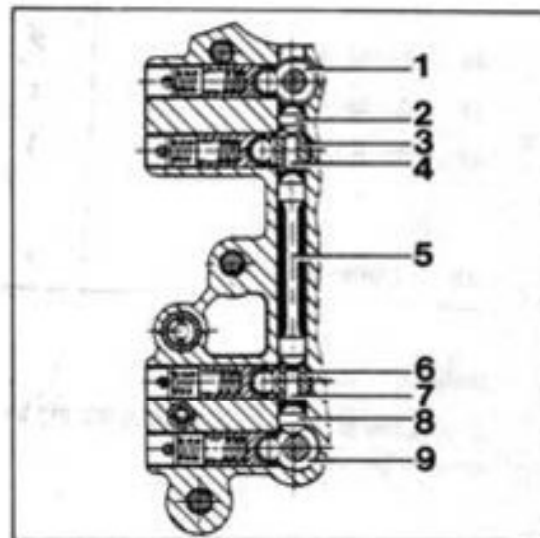
1. The number of adjusting shims "S3" noted during dismantling or the number of adjusting shims noted when the drive pinion was adjusted should be placed onto the studs in the housing until the equivalent thickness is obtained.



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Note

Do not move shift rods across the neutral or gear stop position as this may cause the small intermediate locks to drop out inadvertently.

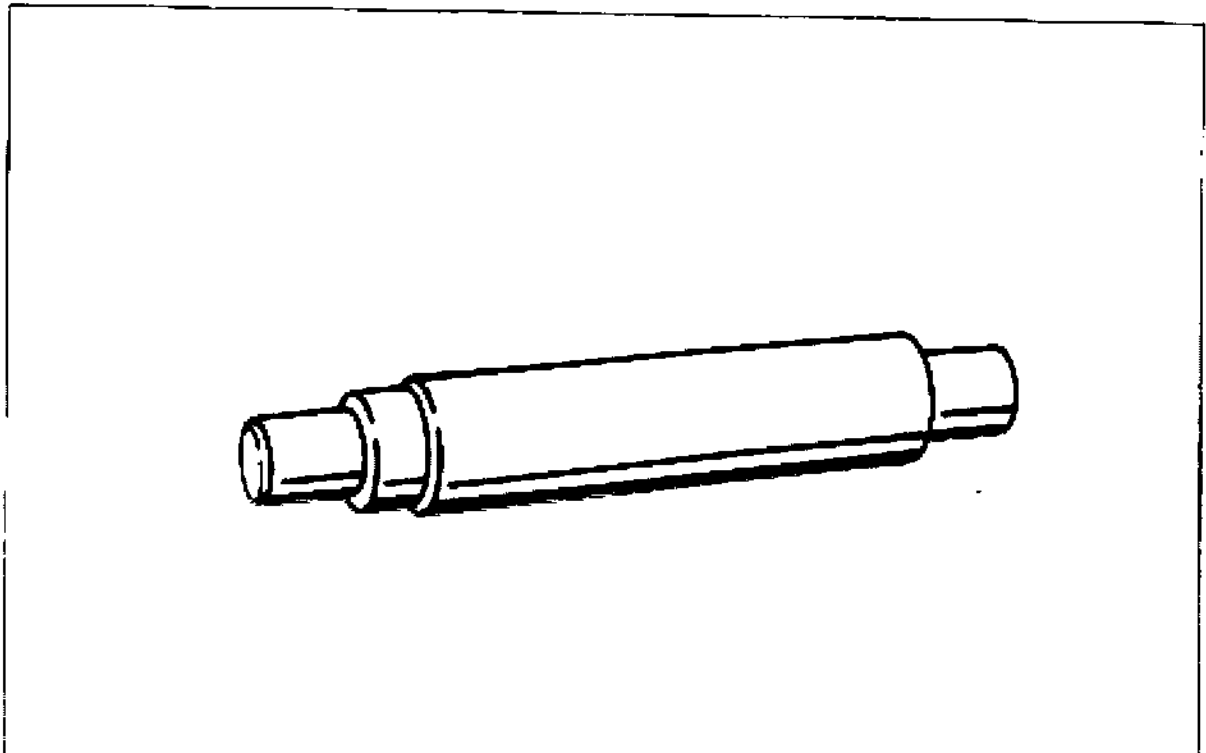


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4 and 7 = Intermediate locks

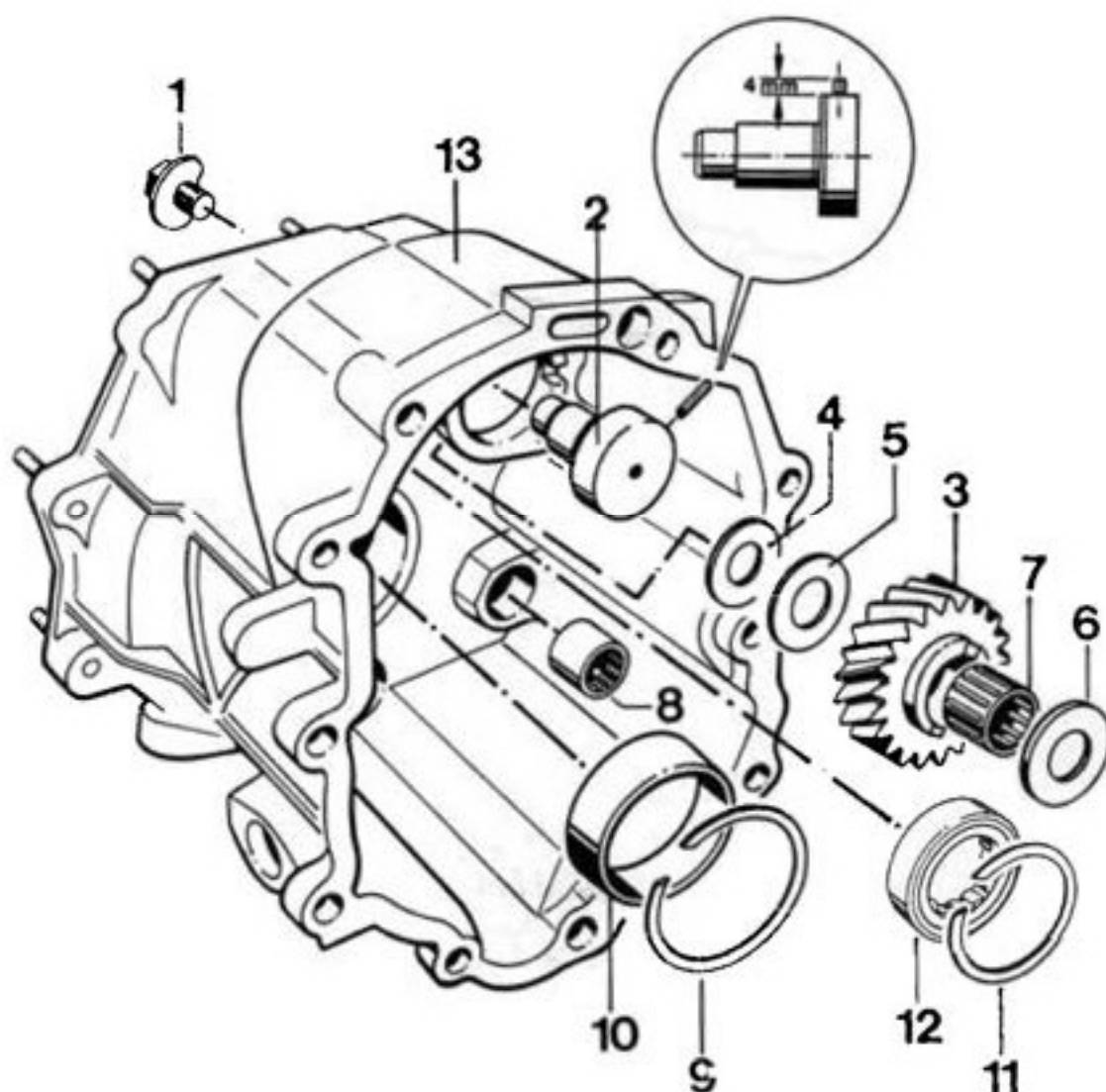
2. Engage 4th gear and assemble fully preassembled gear set to internal selector rod.
3. Tighten tensioning plate lock nuts to 23 Nm (17 ftlb).

34 50 37 Dismantling and assembling intermediate casing
tools



No.	Designation	Special tool	Order number	Explanation
	Mandrel	9254	000.721.925.40	

Dismantling and assembling intermediate casing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt with washer	1		$M_A = 23 \text{ Nm (17 lbf.)}$
2	Pin with roll pin	1		Install in correct position
3	Reverse idler gear	1		Small flange faces hexagon head bolt (item No. 1)
4	Thrust washer (2.0 mm thick)	1		
5	Thrust washer (1.5 mm thick)	1		Do not confuse with washer No. 4
6	Thrust washer (1.5 mm thick)	1		Do not confuse with washer No. 4
7	Needle roller assembly	1		
8	Ball sleeve	1	Push out with suitable drift or a length of pipe (e.g. VW 423)	Working from inside, press in flush across lettered side using 9254
9	Snap ring	1		
10	Bearing outer race	1	Heat housing to approx 120 °C	Heat housing to approx. 120 °C
11	Snap ring	1		
12	Cylindrical roller bearing	1	Heat housing to approx 120 °C and pull out with suitable internal puller (e.g. Schrem 30 - 40)	As No. 10
13	Housing	1		

Notes on assembly

1. Reassemble reverse idler gear with thrust washers and bolt in correct position (refer to exploded view). With thrust washers and bolt located correctly, the end clearance of the reverse idler gear is 0.15...0.35 mm.

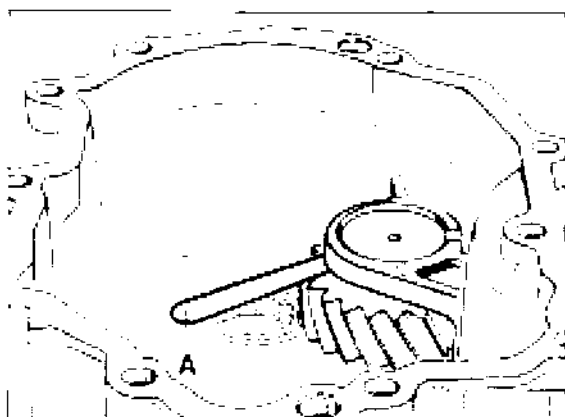


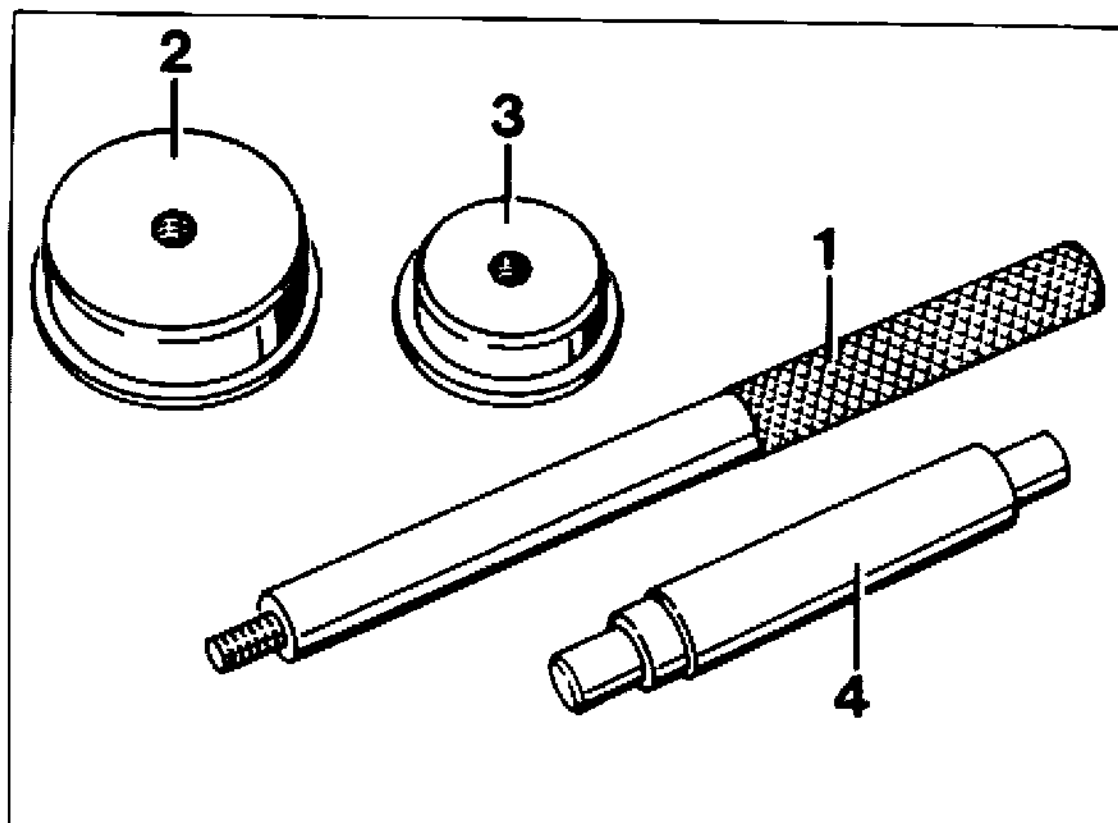
Fig. 34

A = Feeler gauge

2. Check end clearance of reverse idler gear with feeler gauge.

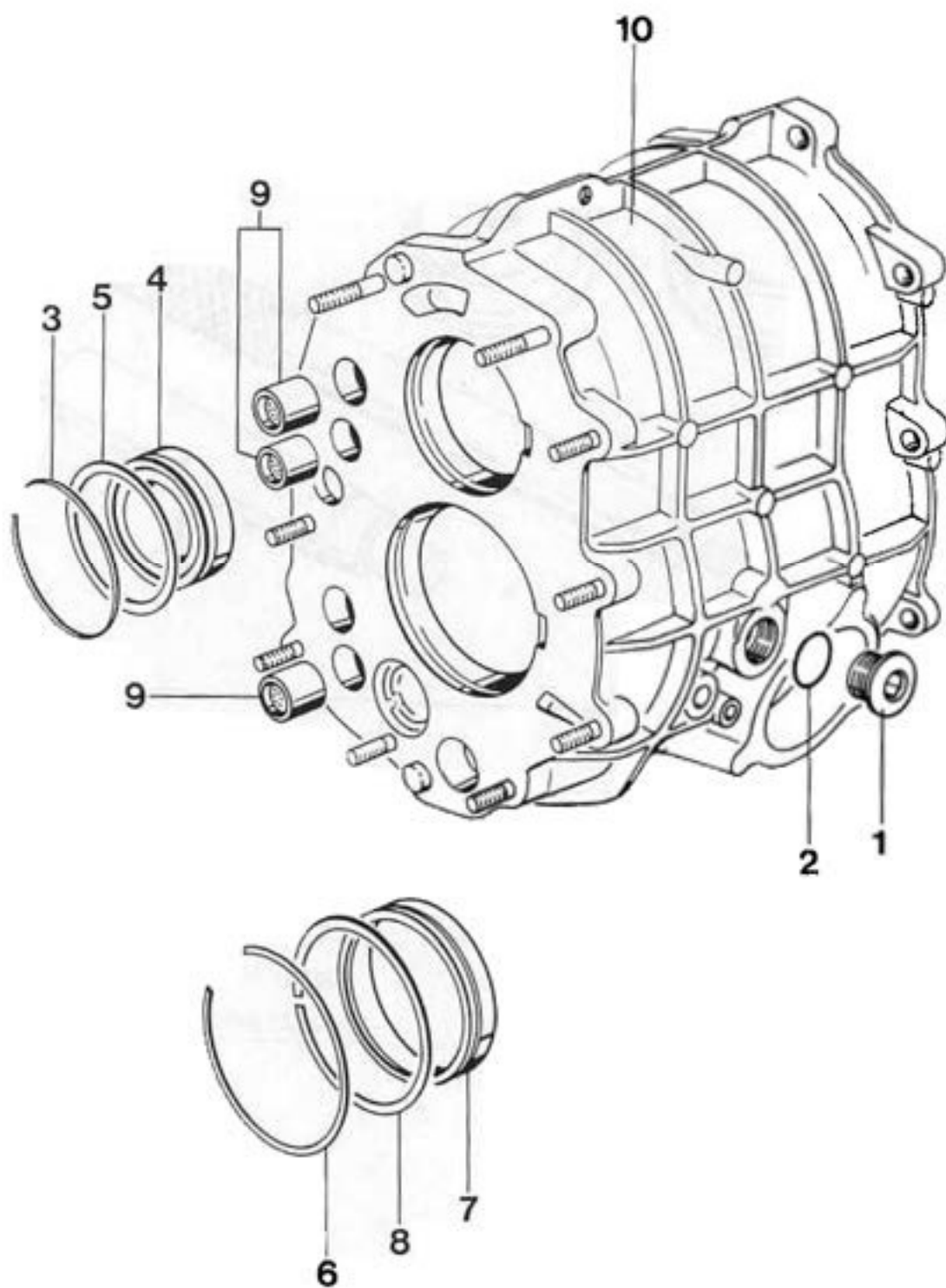
34 52 37 Dismantling and assembling gear housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 a	000.721.254.10	
3	Thrust piece	P 254 b	000.721.254.20	
4	Mandrel	9254	000.721.925.40	

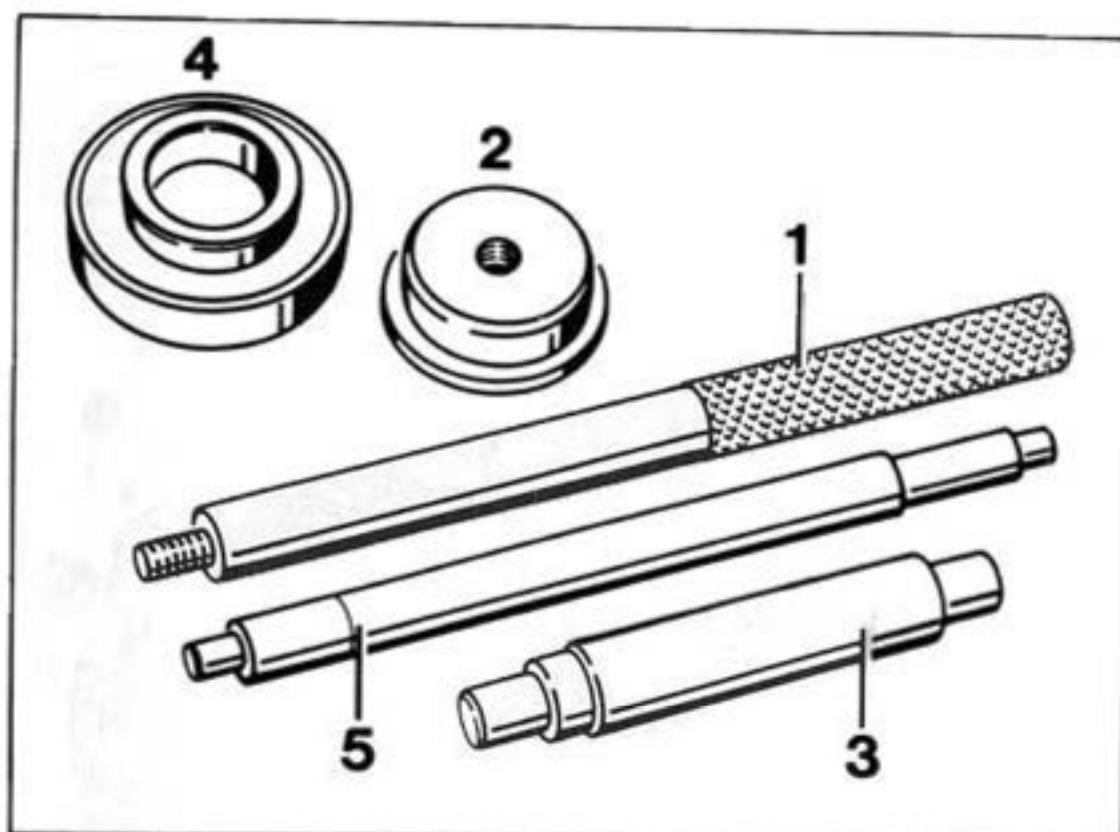
34 52 37 Dismantling and assembling gear housing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Torque: 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Retaining ring	1		
4	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
5	Snap ring	1		
6	Retaining ring	1		
7	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
8	Snap ring	1		
9	Ball sleeve	3	Press out with suitable mandrel	Press in flush with Special Tool 9254
10	Gear housing	1		

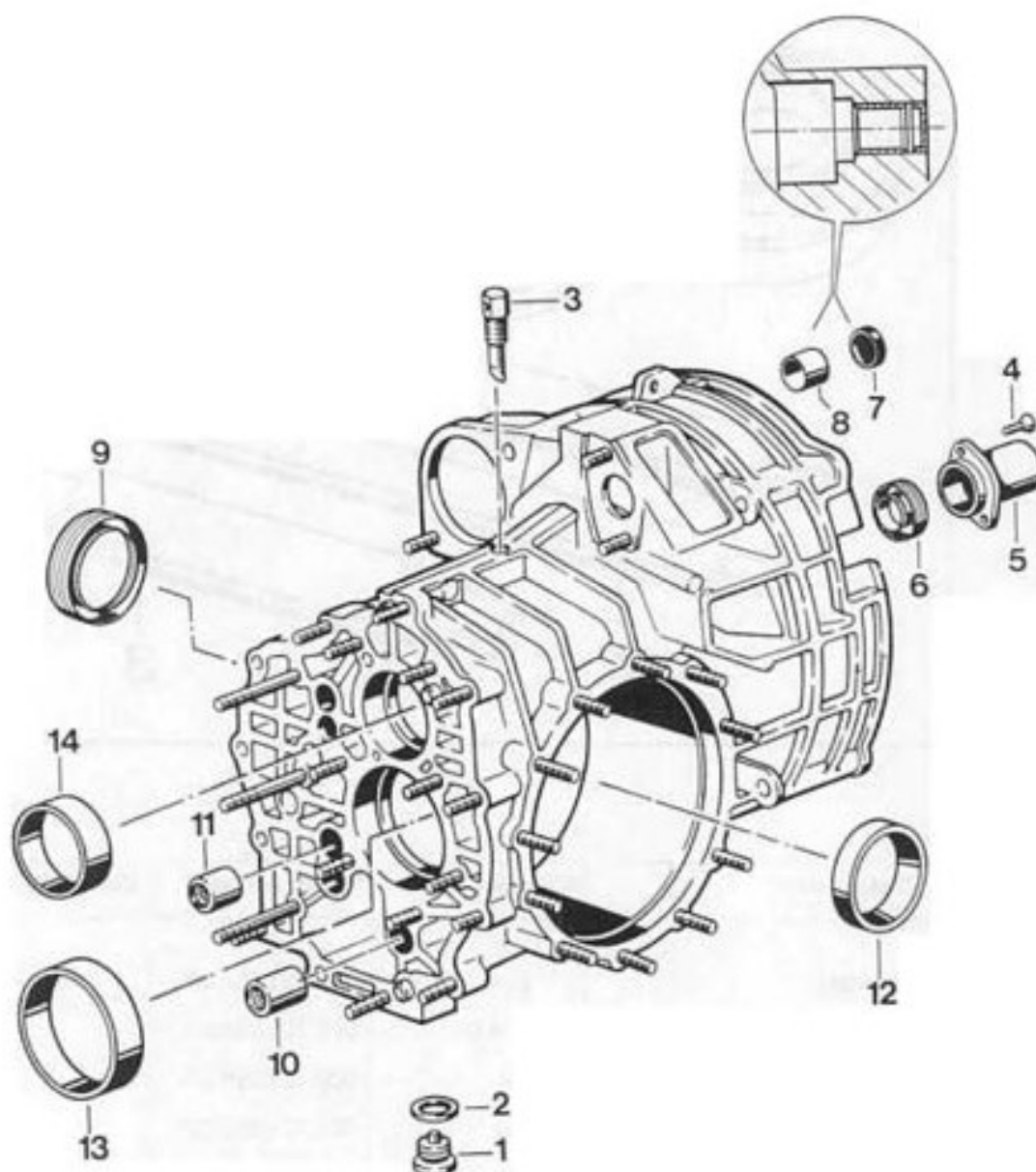
34 37 37 Dismantling and assembling transmission housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 b	000.721.254.20	
3	Mandrel	9254	000.721.925.40	
4	Thrust piece	9252	000.721.925.20	
5	Mandrel	9515	000.721.951.50	

34 37 37 Dismantling and assembling transmission housing

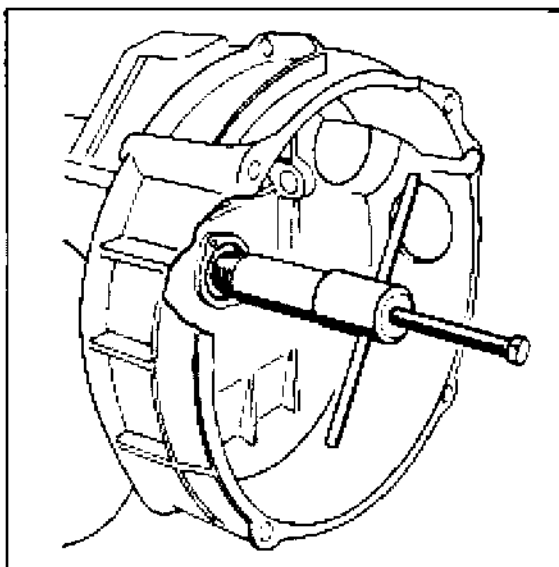


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Clean, tighten to 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Vent	1		Tighten to 35 Nm (26 ftlb.). The hole in the hexagon head must face the transfer casing
4	Oval-head screw	2		Tighten to 10 Nm (7 ftlb.)
5	Guide tube	1		
6	Shaft seal	1		Do not fit until the gear set has been fitted (also refer to page 35 - 201)
7	Cover	1		
8	Bushing	1	Push out with drift 9515, working from inside towards outside	Do not grease or oil and do not clean with solvents (e.g. benzene). Push in from inside with drift 9515 until it is seated against the stop.
9	Shaft seal	1		Push in with Special Tool 9252 until it is seated against the stop. Pace space between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53)
10	Ball sleeve (long)	1	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254
11	Ball sleeve (short)	4	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254

No.	Designation	Qty.	Note:	
			Removal	Installation
12	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable drift	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece
13	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable thrust piece (e.g. VW 513)	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece (e.g. 9247/4) until it is seated against the stop
14	Bearing outer race	1	Heat transmission housing to approx. 120 °C and pull out with suitable internal puller (e.g. Schrem 50 - 60)	Heat transmission housing to approx. 120 °C and press in with Special Tools P 254 and P 254 b until it is seated against the stop
15	Transmission housing	1		

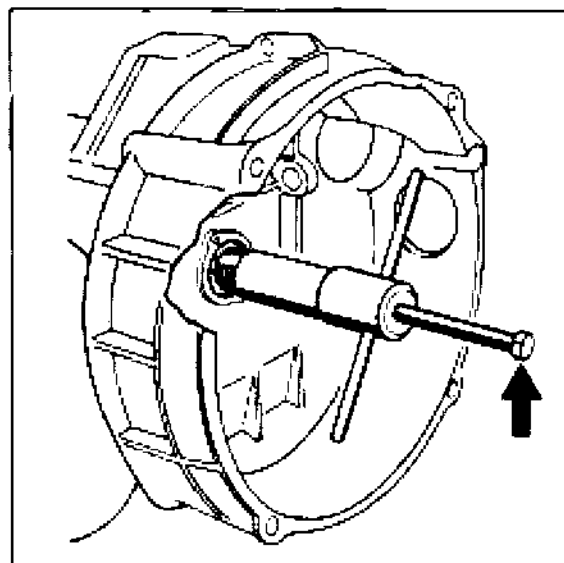
35 50 19 Removing and installing input shaft oil seal**Removing**

1. Remove engine/transmission assembly and separate transmission from engine.
2. Remove release bearing guide tube.
3. Screw Special Tool 9251 firmly into the oil seal.



1652-34

4. Screw in the hexagon-head bolt and pull out oil seal.



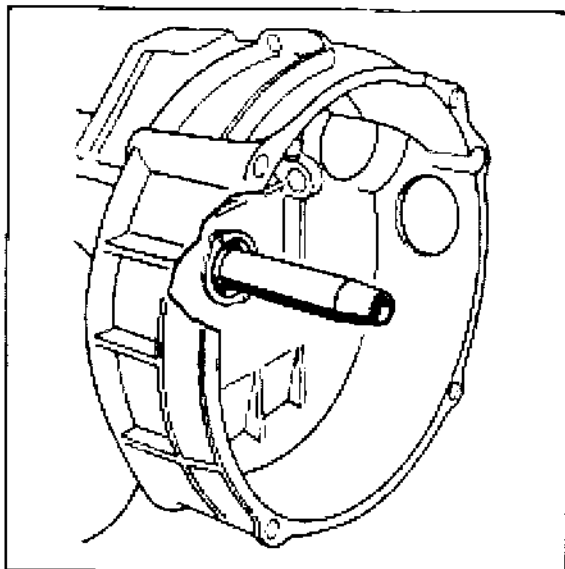
1653-34

Note

If the helical tension spring jumps out when removing the seal, use a wire hook to pull it off the input shaft.

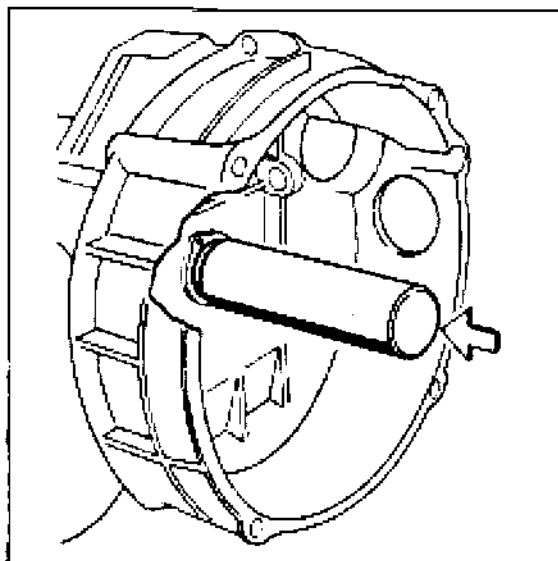
Installing

1. Push assembling sleeve 9255 onto the input shaft splines.



1654-34

3. Use Special Tool 9256 to drive the oil seal into place.

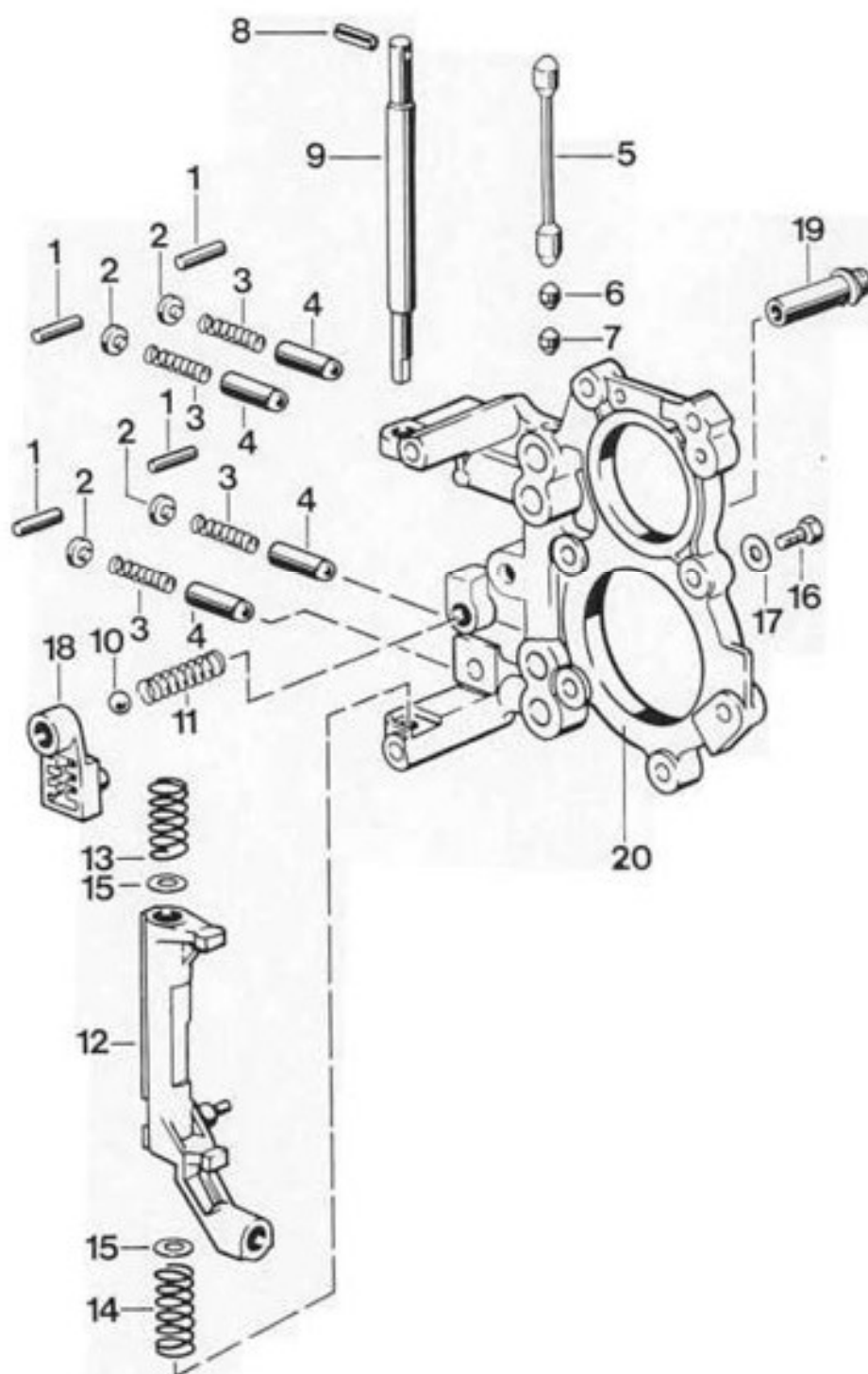


1655-34

2. Pack oil seal area between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53).

4. Install guide tube and tighten mounting screws to **10 Nm (7 flb.)**.

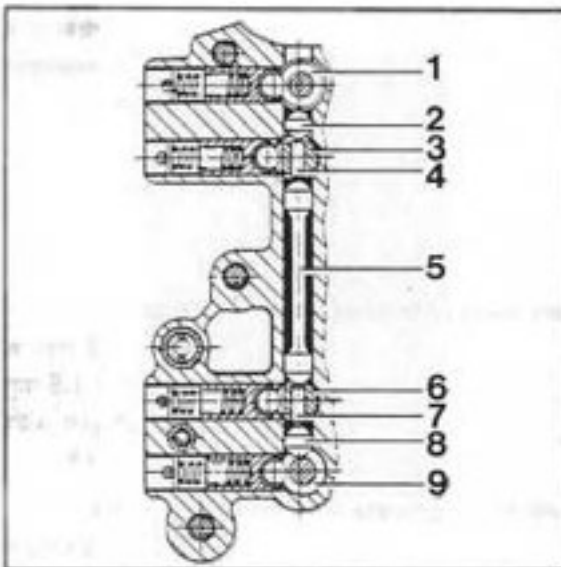
35 19 37 Dismantling and assembling tensioning plate



No.	Designation	Qty.	Removal	Note: Installation
1	Straight pin	4	With the shift rods fitted, the springs (No. 3) are under load	
2	Washer	4		
3	Thrust spring	4		
4	Locking bush	4		
5	Lock (long)	1		
6	Lock (short)	1		
7	Lock (short)	1		
8	Roll pin	1		
9	Bearing shaft	1		Install in correct position, use 8 mm dia. locating mandrel to locate relative to tensioning plate hole
10	Ball	1		
11	Thrust spring	1		
12	Deflection lever	1		
13	Thrust spring	1	Mark for reinstallation	Free length = 51.3 ± 0.5 mm wire thickness = 1.5 mm. Do not confuse with spring No. 14
14	Thrust spring	1	Mark for reinstallation	Free length = 72.6 ± 0.5 mm wire thickness = 1.1 mm. Do not confuse with spring No. 13
15	Shim	2		

No.	Designation	Qty.	Note:	
			Removal	Installation
16	Hexagon head bolt	1	Press out with suitable drift	Tighten to 10 Nm (7 ftlb.)
17	Washer	1		
18	Selector gate	1		
19	Adapter sleeve with snap ring	1		Press in with suitable drift until it is seated against the stop
20	Tensioning plate	1		

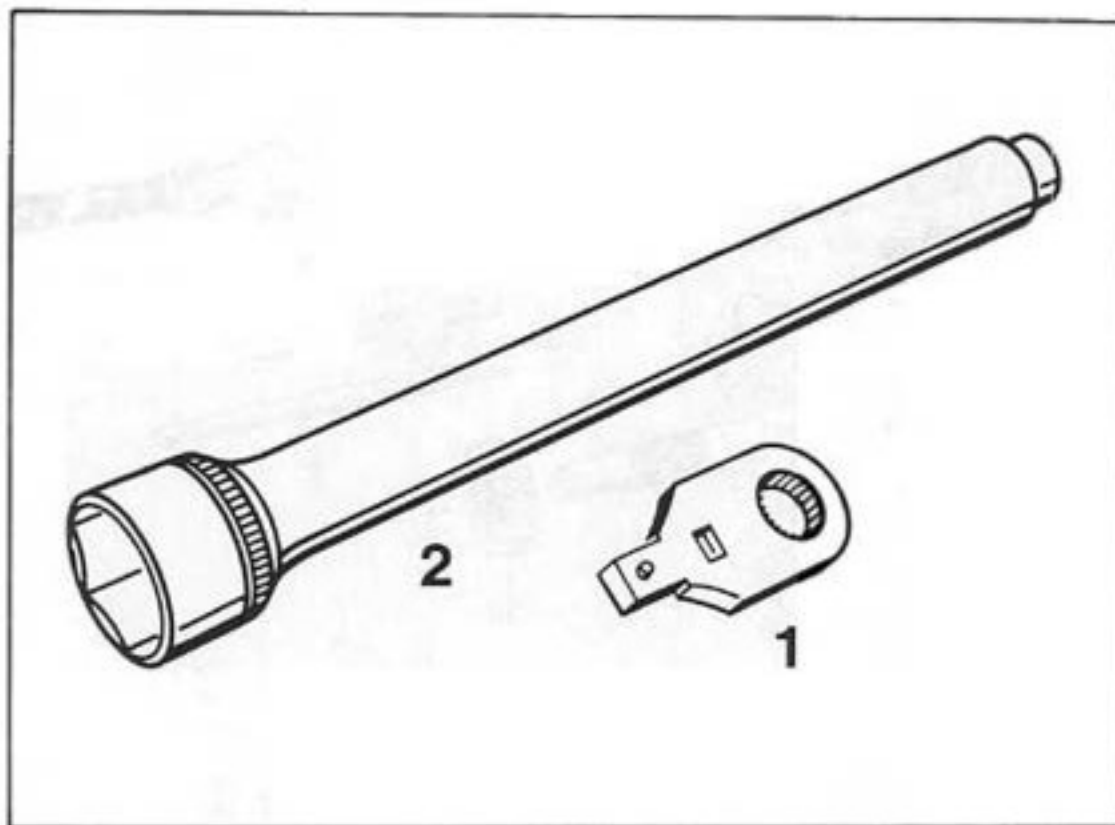
Installation position of the locks



1709-35

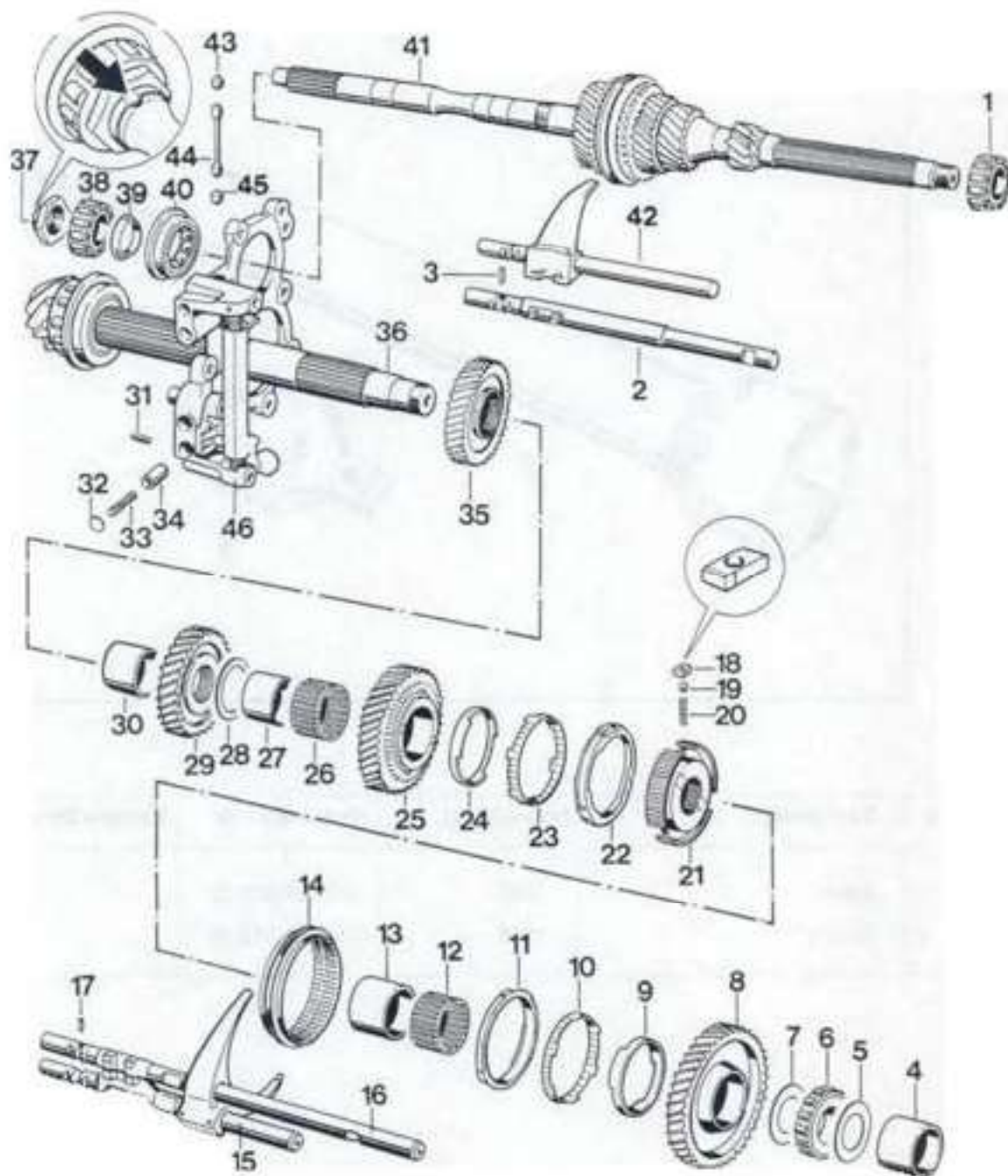
35 40 19 Removing and installing input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 19 Removing and installing input shaft



No.	Designation	Qty.	Removal	Note:	
				Installation	
1	Cylindrical roller bearing	1		Heat to approx. 120 °C	
2	Shift rod (5th and 6th gear)	1	Shift rods in neutral position		
3	Spacer	1		Coat with stiff grease to insert	
4	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C	
5	Thrust washer	1			
6	Cylindrical roller bearing	1		Heat to approx. 120 °C	
7	Thrust washer	1			
8	Loose gearwheel (1st gear)	1			
9	Friction ring	1		Fit with the same gear-wheel	
10	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel	
11	Synchronizing ring	1		Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tapered ring. Three lugs face the driver dogs.	
12	Needle roller bearing	1		Fit with the same gear-wheel	
13	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C	

No.	Designation	Qty.	Note:	
			Removal	Installation
14	Shift sleeve	1		Insert complete with guide sleeve and shift rods. Make sure the missing tooth of the internal teeth of the guide sleeve (No. 21) is aligned exactly above the oil bore of the output shaft. The circumferential identification groove must face 2nd gear. The centerpunch marks must be centered relative to the balls (see page 35 - 213)
15	Shift rod with pinned shift fork	1		
16	Shift rod (reverse)	1		
17	Intermediate lock	1		Coat with stiff grease to insert
18	Driver dog	3		Install in correct position
19	Ball	3		
20	Spring	3		
21	Guide sleeve	1		Missing tooth of the internal splines must be exactly above the oil bore of the output shaft
22	Synchronizing ring	1		Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered sleeve. Three lugs face the driver dogs.

No.	Designation	Qty.	Note:	
			Removal	Installation
23	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
24	Friction ring	1		Fit with the same gear-wheel
25	Loose gearwheel (2nd gear)	1		
26	Needle roller bearing	1		Fit with the same gear-wheel
27	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C
28	Thrust washer	1		
29	Fixed gearwheel (3rd gear)	1		Large collar faces thrust washer No. 28
30	Spacer sleeve	1		
31	Straight pin	4		
32	Washer	4		Guide lug faces spring
33	Thrust spring	4		
34	Latch	4		
35	Fixed gearwheel (4th gear)	1		Large collar faces four-point bearing
36	Output shaft	1		
37	Flange nut*	1	Undo with Special Tools 9282 and 9105	Tighten to 250 Nm (184 ftlb.). Upset the flange to lock
38	Cylindrical roller bearing	1	Press off across 2nd gear fixed gearwheel	Heat to approx. 120 °C
39	Bearing inner race	1		Heat to approx. 120 °C

No.	Designation	Qty.	Note:	
			Removal	Installation
40	Four-point bearing	1		
41	Output shaft	1		Insert complete with pinned shift rod / shift fork (No. 42)
42	Shift rod with pinned shift fork	1		
43	Lock (short)	1		
44	Lock (long)	1		
45	Lock (short)	1		
46	Tensioning plate	1		

Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

Removal and installing notes

Removal

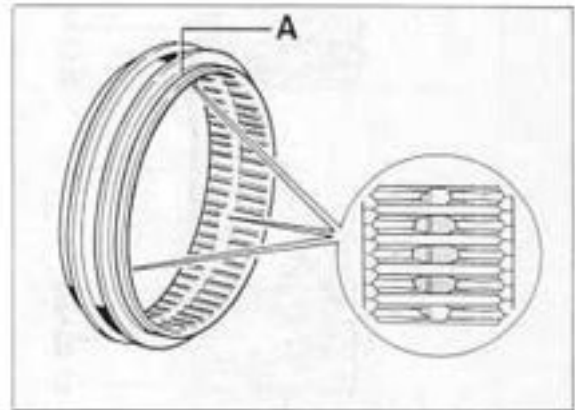
Note

Parts No. 1 to 30 are removed with the tensioning plate remaining fitted (refer to page 34 - 209).

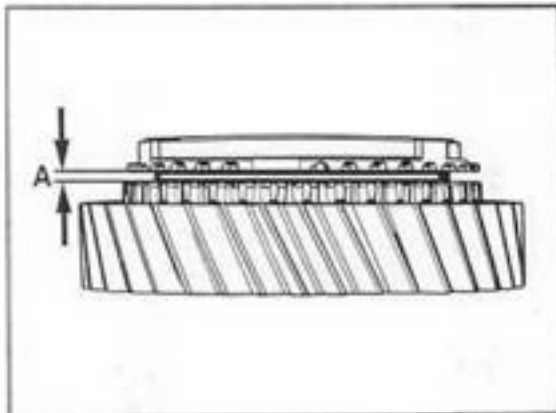
Installation

1. Using a suitable flat iron bar, clamp tensioning plate in a vise in such a manner that the hole for the shift rod locks is horizontal.
2. Check synchromesh of 1st and 2nd gears. To do so, place friction ring, tapered ring and synchronizing ring in correct position onto gearwheel. Check gap "A" with a feeler gauge.

Installation dimension (new) = 1.5 to 2.0 mm
Wear limit = 1.2 mm



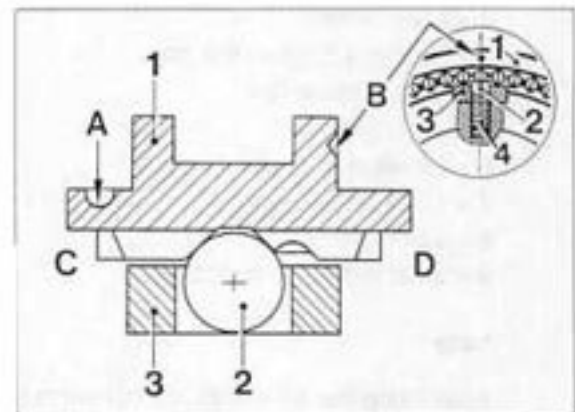
A = Identification groove (must face 2nd gear)



1701-35

3. Place shift sleeve onto guide sleeve so that centerpunch marks "B" are centered relative to the balls.

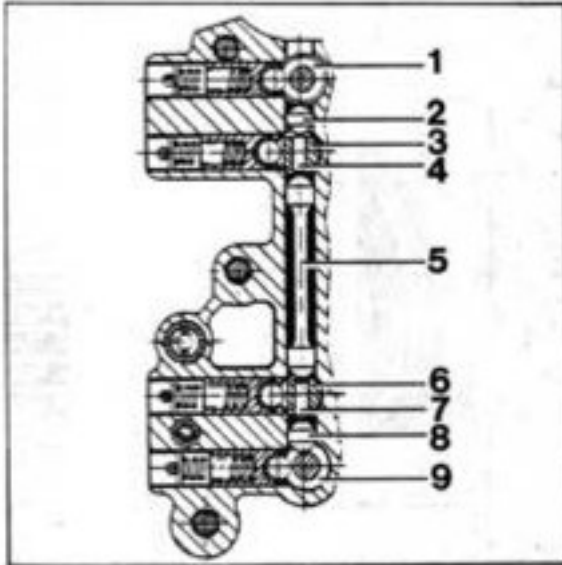
In addition, the circumferential identification groove "A" must face 2nd gear.



1906-35

- 1 = Shift sleeve
- 2 = Ball
- 3 = Driver dog
- 4 = Spring
- A = Identification groove
- B = Centerpunch mark
- C = 2nd gear side
- D = 1st gear side

4. Observe installation position of locks.



1799-35

- 1 = Shift rod 3rd and 4th gear
- 2 = Lock (short)
- 3 = Shift rod 5th and 6th gear
- 4 = Intermediate lock
- 5 = Lock (long)
- 6 = Reverse shift rod
- 7 = Intermediate lock
- 8 = Lock (short)
- 9 = Shift rod 1st and 2nd gear

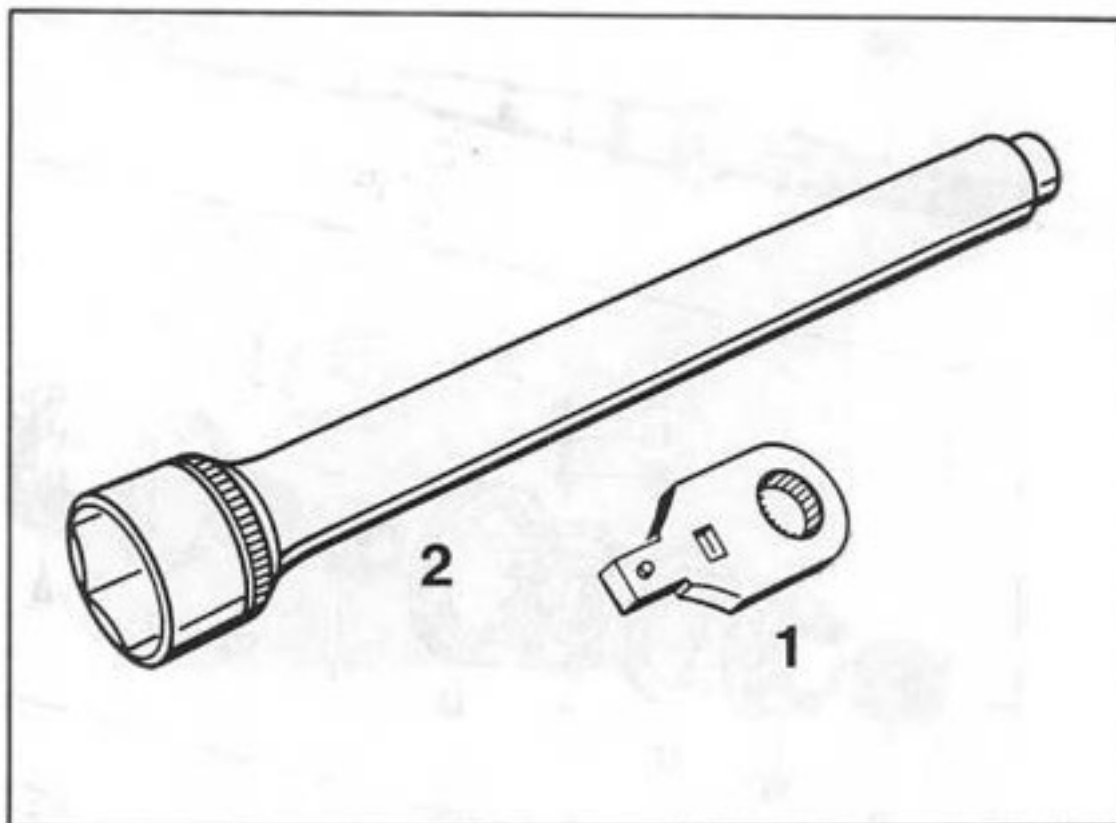
Note

After fitting the shift rods, do not move them across the neutral or gear latch positions as this may cause the small intermediate locks to drop out inadvertently.

To avoid inadvertent movement of the shift rods, lock shift rods by engaging 3th gear.

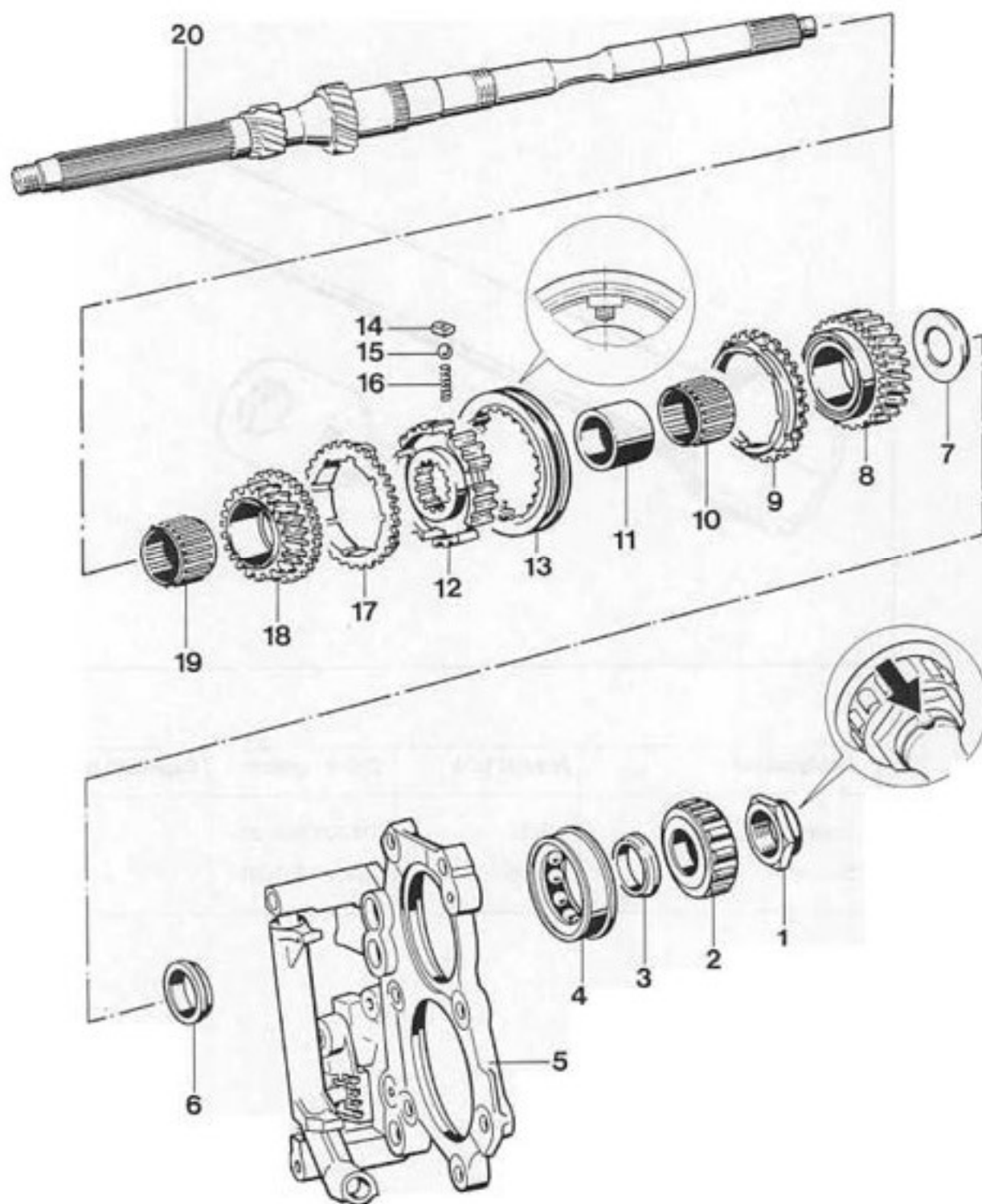
35 40 37 Dismantling and assembling input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 37 Dismantling and assembling input shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon nut*	1	Undo with Special Tools 9282 and 9105	Use Special Tools 9282 and 9105 and tighten to 250 Nm (184 ftlb.). Up-set flange to lock
2	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
3	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
4	Four-point bearing	1		
5	Tensioning plate	1		Clamp in a vise (use protective jaws), insert lock (long), engage 5th gear and install input shaft with pinned shift rod/shift fork
6	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1)	Heat to approx. 120 °C
7	Thrust washer	1		Large, face-ground side faces needle cage
8	Loose gearwheel (4th gear)	1		Replace only in pairs
9	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position using the same gearwheel (lugs face the driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
10	Needle-roller assembly	1	Mark cage	Install with the same gear-wheel
11	Inner race	1		Heat to approx. 120 °C
12	Guide sleeve	1	Remove complete with shift sleeve	Install complete with shaft sleeve and synchromesh components
13	Shift sleeve (3rd and 4th gear)	1	Make sure synchromesh components do not pop out.	Observe installation position, install complete with guide sleeve and synchromesh components. Center the center-punch marks relative to the driver dogs
14	Driver dogs	3		Install in correct position, domed side faces shift sleeve
15	Ball	3		
16	Spring	3		
17	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position with the same gearwheel (lugs face the driver dogs)
18	Loose gearwheel (3rd gear)	1		Replace only in pairs
19	Needle-roller assembly	1	Mark for reinstallation	Fit with the same gear-wheel
20	Drive shaft	1		

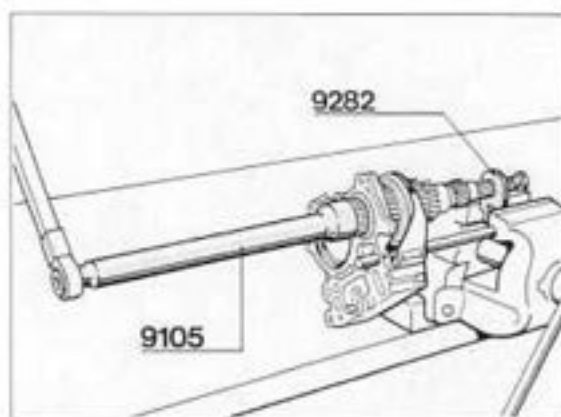
* Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every repair.

Dismantling and assembling notes

Dismantling

1. Mount retaining plate 9282 in vise, fit input shaft and undo hexagon-head nut with Special Tool 9105.



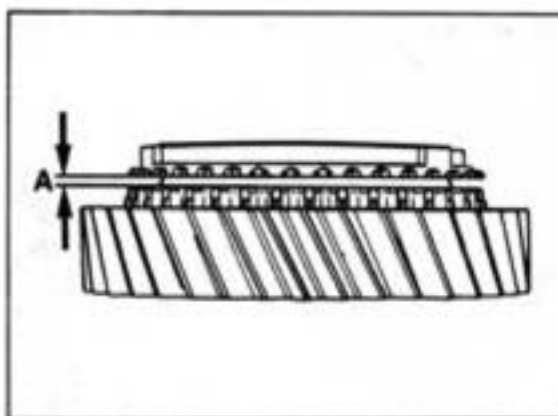
535-35

2. Use a suitable tool (e.g. Kukko 17 - 1) to press all parts off the input shaft over the 2nd gear wheel.

Assembling

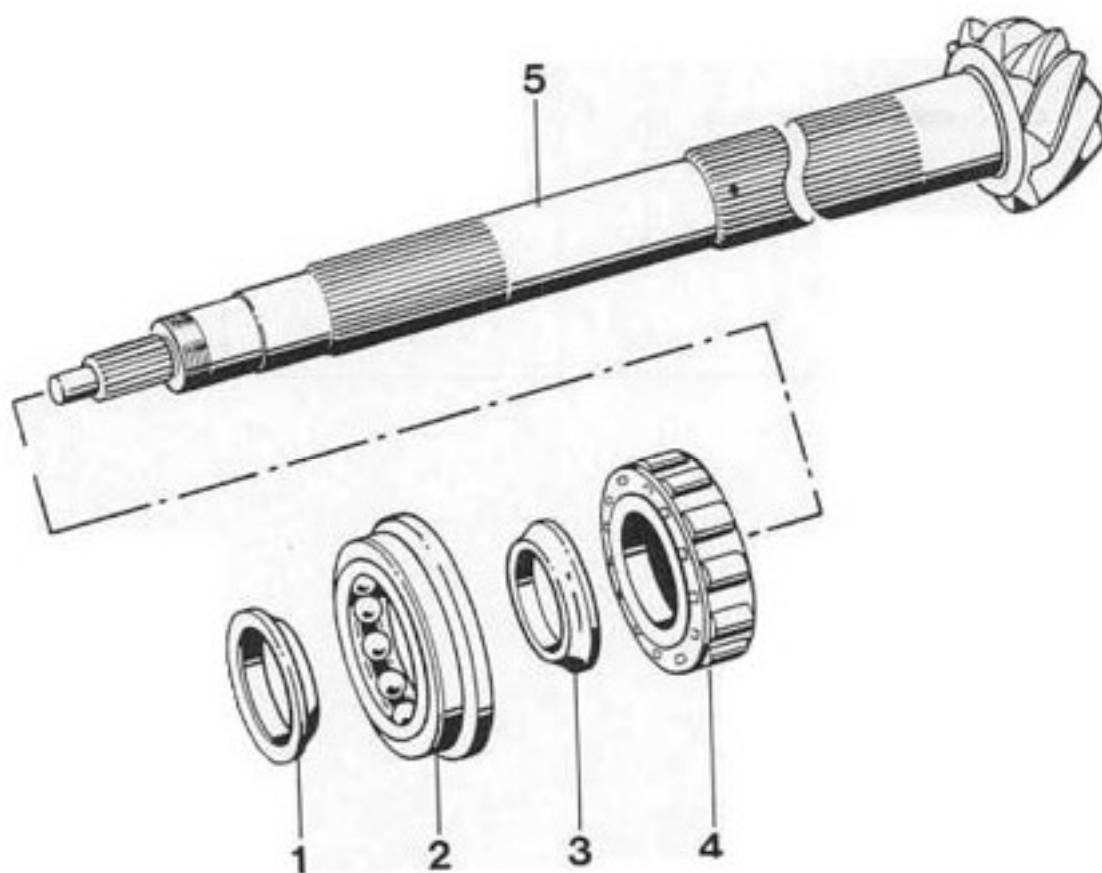
1. To inspect synchronizing rings, press rings onto the gear wheel tapers and use a feeler gauge to measure gap "A".

Assembly dimension (new) = 0.9 mm min.
Wear limit = 0.6 to 0.7 mm



518-35

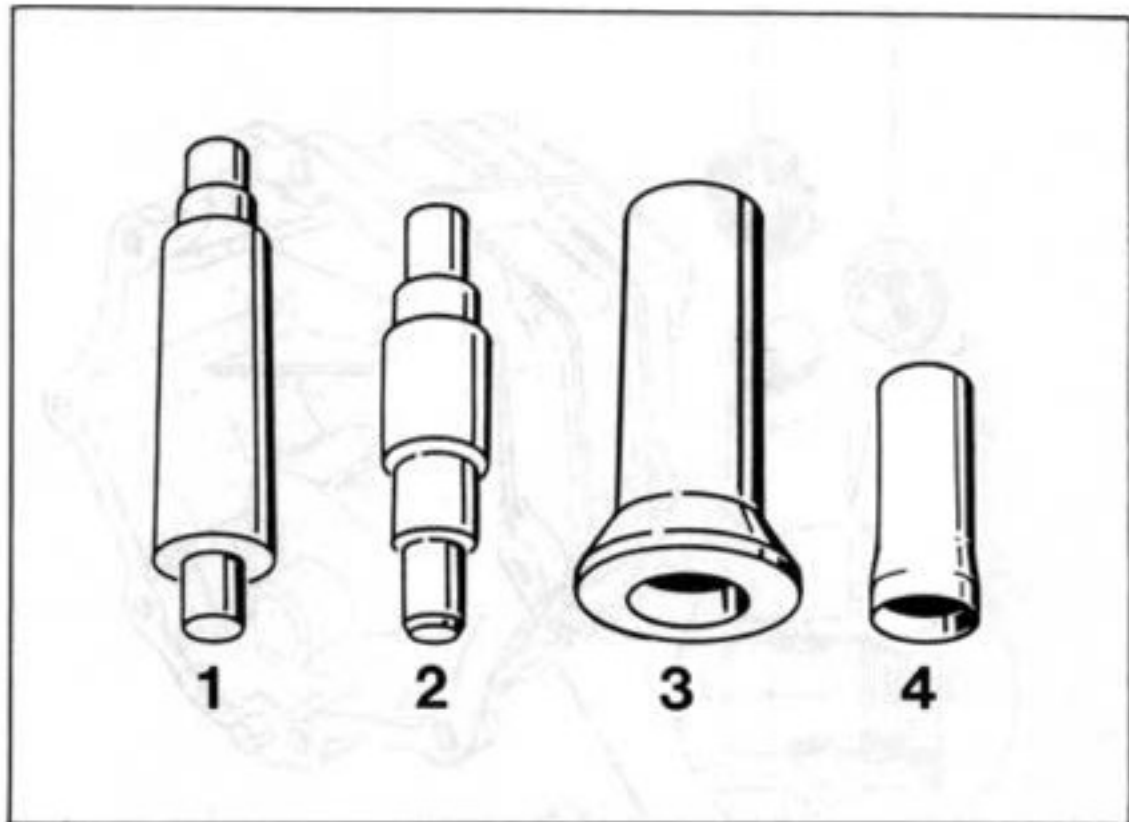
35 59 37 Dismantling and assembling output shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 3, heat to approx. 120 °C and press into place
2	Four-point bearing	1		
3	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 1, heat to approx. 120 °C and press into place
4	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 15 - 17)	Heat to approx. 120 °C and press into correct position using a suitable pipe section (e.g. VW 519)
5	Output shaft	1		Observe matching number. Readjust if required

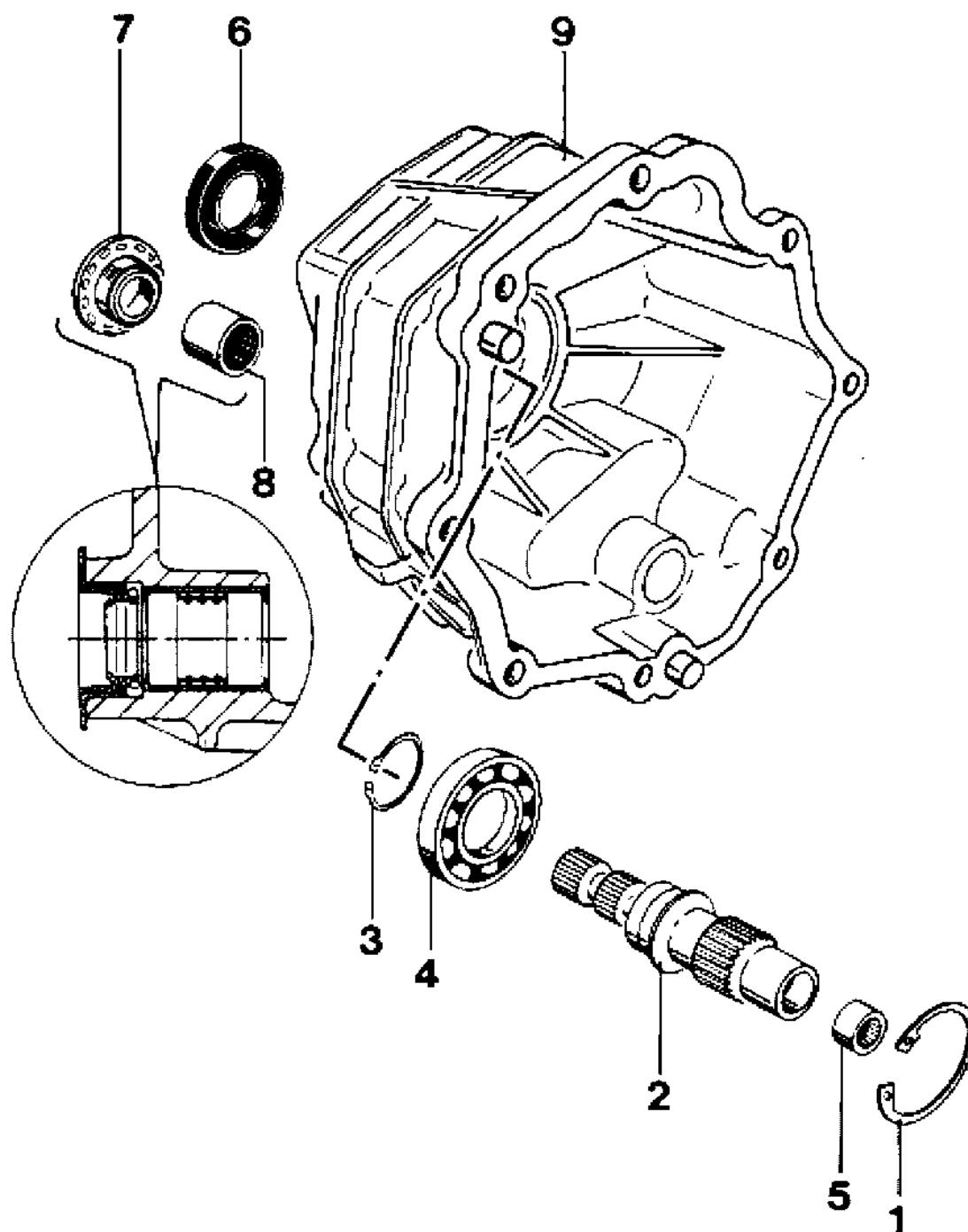
39 66 37 Dismantling and assembling transfer casing

Tools



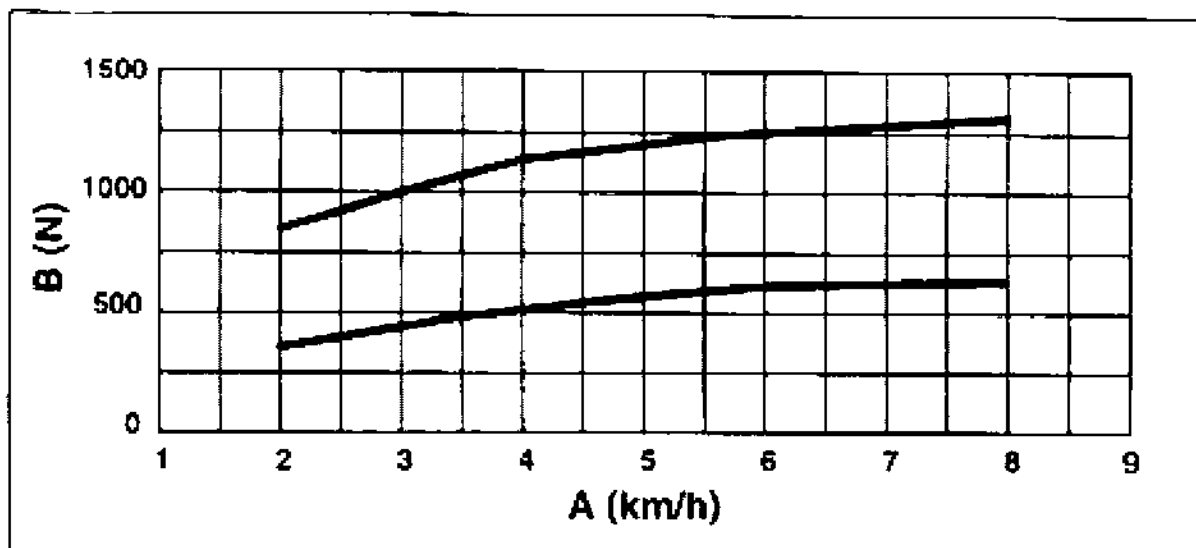
No.	Designation	Special tool	Order number	Explanation
1	Mandrel	9254	000.721.925.40	
2	Mandrel	9223	000.721.922.30	
3	Thrust piece	9234	000.721.923.40	
4	Assembly sleeve	9575	000.721.957.50	

39 66 37 Dismantling and assembling transfer casing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Snap ring	1		
2	Output shaft	1		
3	Snap ring	1		
4	Deep-groove ball bearing	1	Press off	Heat to approx. 120 °C
5	Needle roller bearing sleeve	1	Pull out with internal puller	Replace, drive in flush with suitable drift (e.g. P 361)
6	Oil seal	1		Replace, drive in with Special Tool 9575 and 9234 after fitting the output shaft
7	Oil seal	1		Replace, pack space between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53), push in to stop with Special Tool 9254
8	Ball sleeve	1	Press out with suitable drift or pipe section (e.g. VW 423)	Replace, drive home into correct position using Special Tool 9223
9	Housing	1		

39 60 01 Checking operation of installed viscous clutch



2019-39

A = Driving speed of front axle in kilometers per hour

B = Pulling power of front wheels in Newton (N)

Test conditions:

To check, viscous clutch must be at room temperature

Test duration approx. 45 seconds (Determine measurement after 20 seconds)

Repeat measurement after viscous clutch has cooled off for 1 hour

1. Connect system tester and select „Actual values“ menu (ABS speed) (Refer to Vol. VIII, page 45 - 37).
2. Place front wheels of vehicle onto brake dynamometer.
3. Switch off engine, engage parking brake and shift transmission into neutral position.

4. When the rollers have started, switch on ignition and press key > of system tester. The tester will then return to the point where diagnosis was interrupted.

5. Set front wheels into motion with dynamometer rollers (**max. 5 mph**) and determine tractive power of front wheels after 20 seconds.

Note

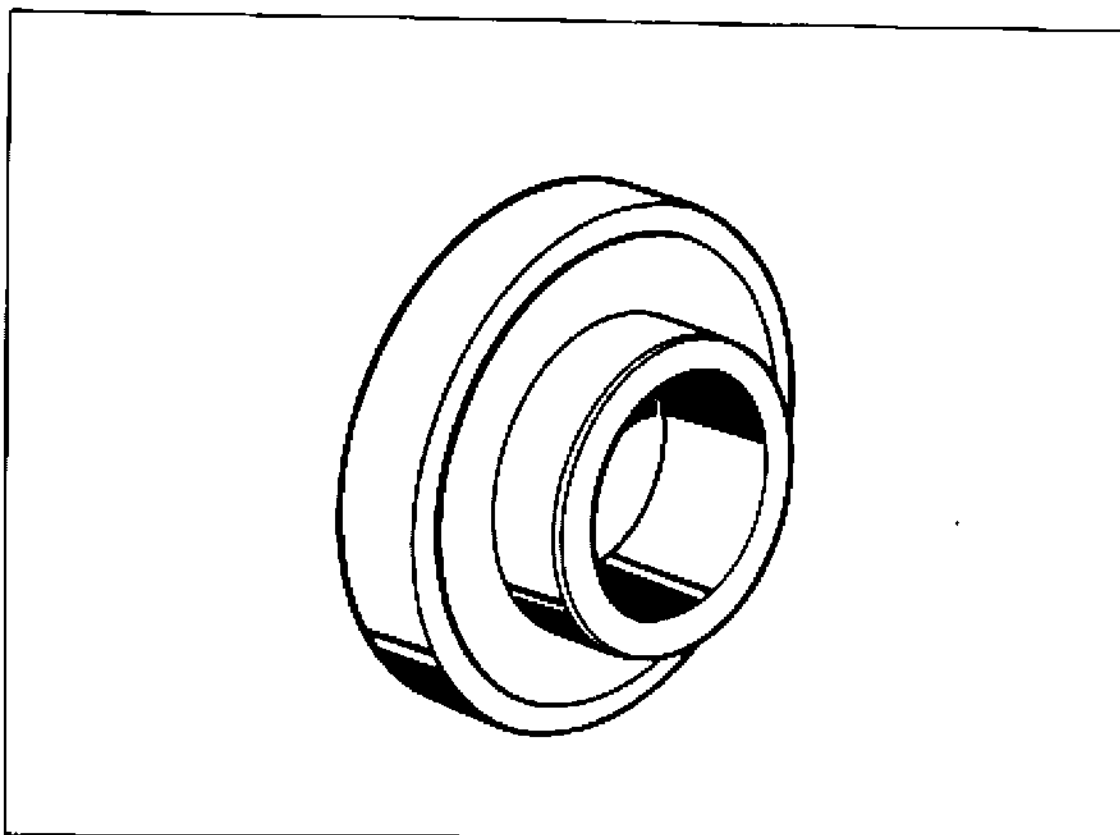
When measuring the wheels individually, add measurements obtained for both front wheels.

5. The viscous clutch is operative if the tractive power of the front wheels is within the boundary curves on the diagram.

The driving speed of the front axle must be taken into account during these measurements (speed can be read off on System Tester 9288).

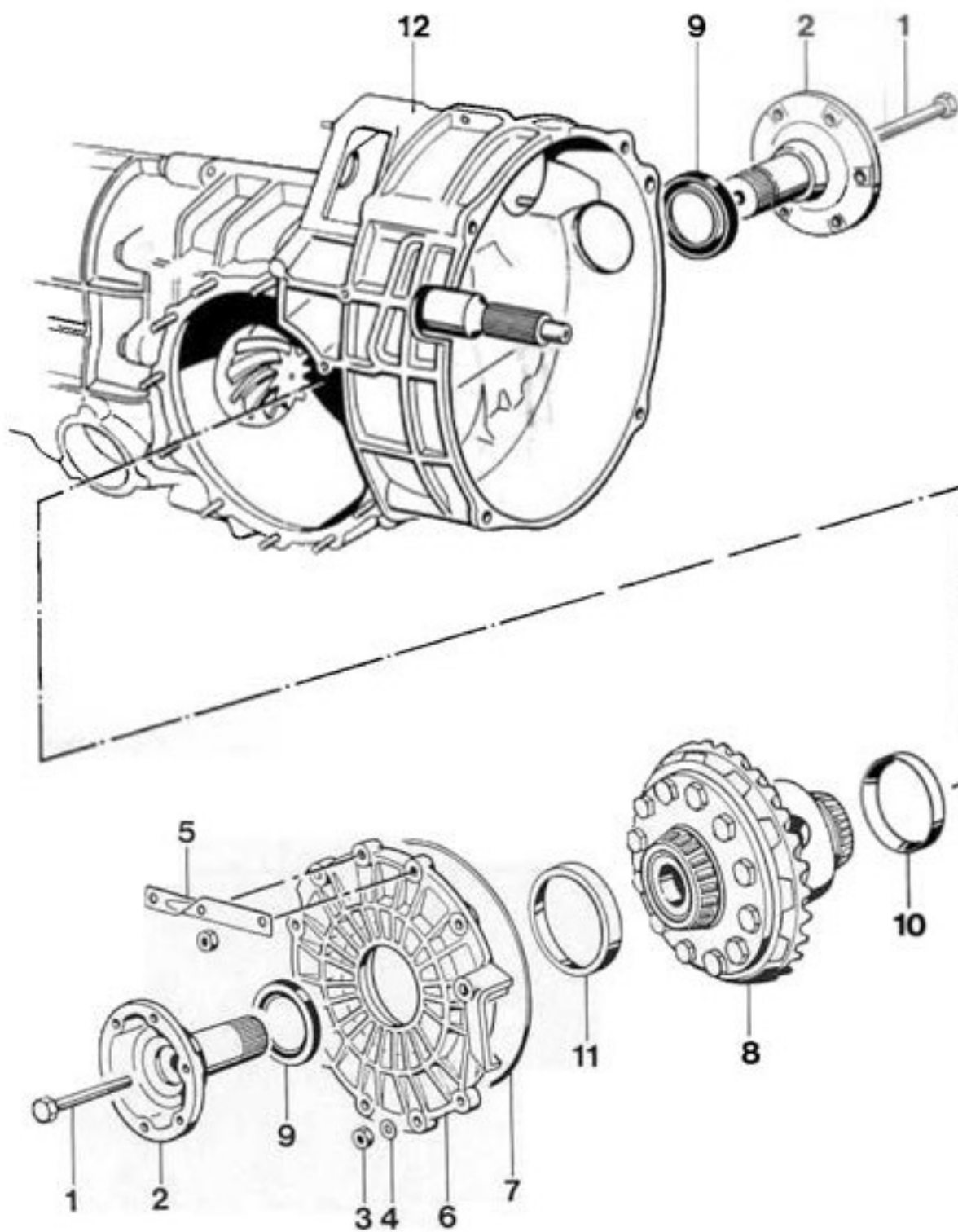
Example:

At a vehicle speed of 3 mph at the front axle, the tractive power must be between approx. 580 N and 1180 N if the viscous clutch is o.k. (cf. diagram).

39 40 19 Removing and installing differential lock**Tools**

No.	Designation	Special tool	Order number	Explanation
	Thrust piece	9252	000.721.925.20	

39 40 19 Removing and installing differential lock

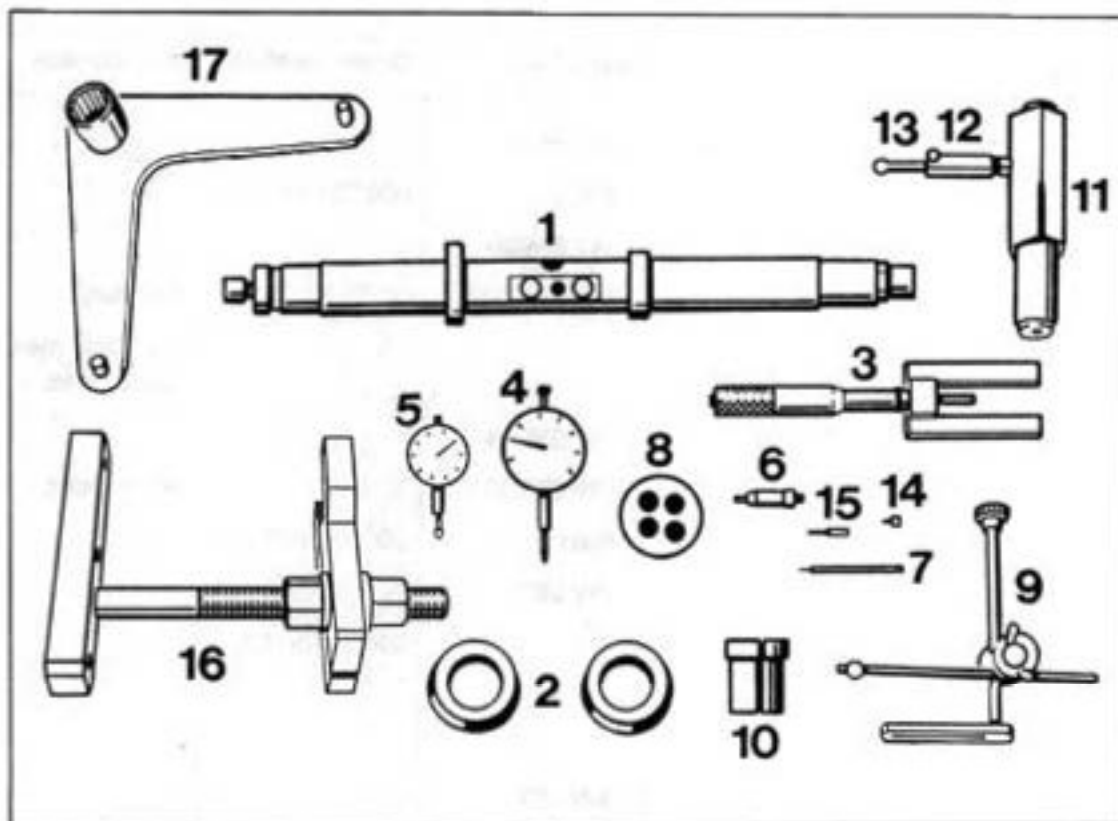


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt	2		Tighten to 44 Nm (32 lbf·ft)
2	Joint flange*	2		
3	Hexagon head nut	11		Tighten to 23 Nm (17 lbf·ft)
4	Washer	11		
5	Holder	1		
6	Cover	1		
7	O-ring	1		Replace, oil lightly, do not twist
8	Differential or limited-slip differential	1		Readjust if required
9	Seal*	2		Pack space between dust and sealing lips with grease (e.g. Liqui Moly Pu 53). Use thrust piece 9252 to press in until it is seated against the stop
10	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 8)	Press in with suitable thrust piece
11	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 9)	Press in with suitable thrust piece
12	gear box	1		

* The halfshaft flange and seal ring can also be removed and installed with the transmission installed.

39 08 15 Adjusting drive set

Tools



39 08 15 Adjusting drive set

Tools

No.	Designation	Special tool	Order number	Explanation
1	Measuring mandrel	VW 385/1		
2	Centering disks	9109	000.721.910.90	
3	Master gauge	VW 385/30		
4	Dial gauge	-		Standard
5	Dial gauge	-		Standard, measuring range 3 mm
6	Gauge plunger	VW 385/14		
7	Dial gauge extension	VW 385/56		30 mm long
8	Gauge block plate	9281	000.721.928.10	
9	Dial gauge bracket	VW 387		
10	Clamping sleeve	9145	000.721.914.50	
11	Adjusting device	VW 521/4		
12	Measuring lever	VW 388		
13	Gauge plunger	VW 388		
14	Dial gauge extension	VW 382/10		6.0 mm long
15	Dial gauge extension	VW 385/15		9.3 mm long
16	Clamping tool*	9577	000.721.957.70	
17	Holder	9253	000.721.925.30	

Practical procedure when readjusting the drive set

If it is necessary to adjust drive pinion and ring gear, follow the below sequence to ensure an efficient working procedure:

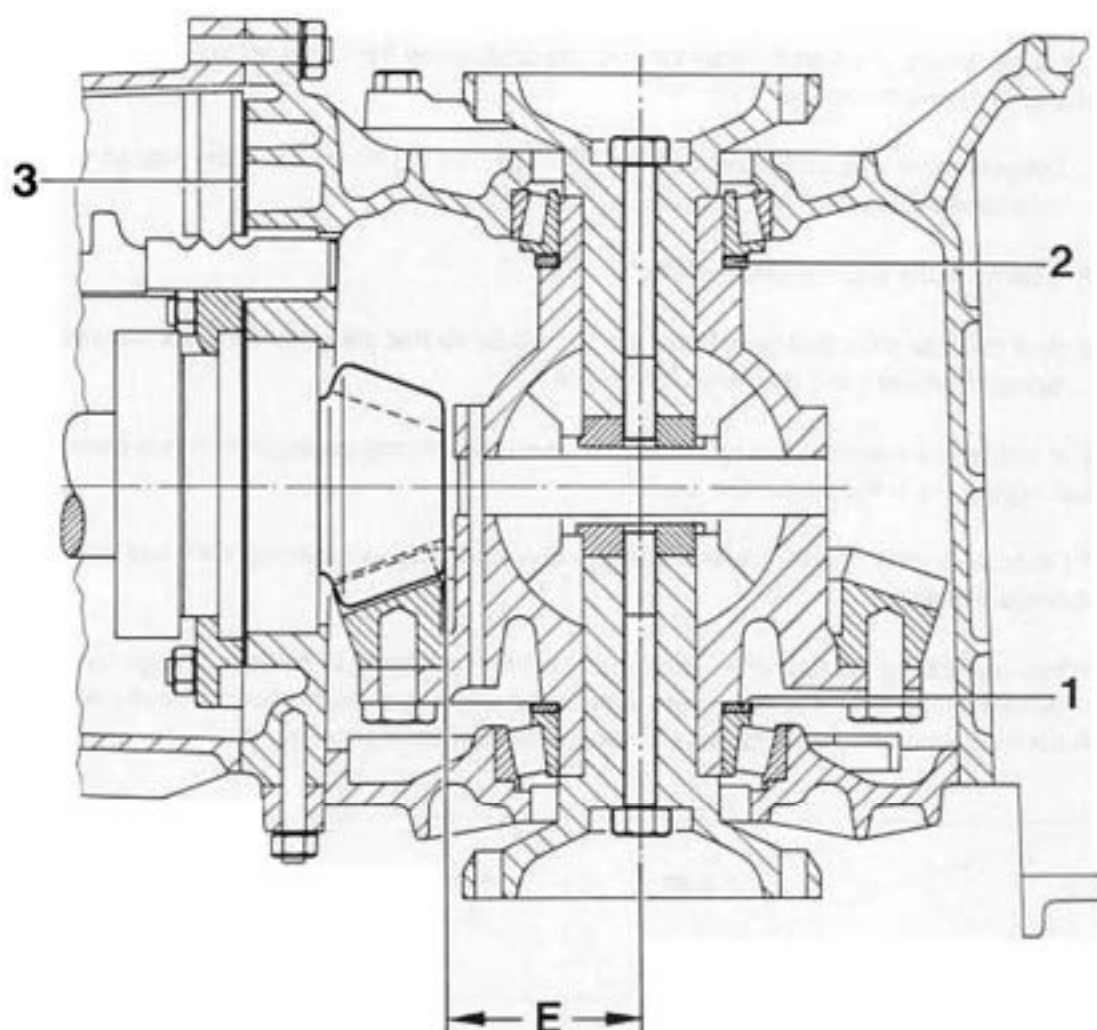
1. Determine the total thickness of shims "Stot" (S_1 plus S_2) for the specified preload on the tapered roller bearings/differential.
2. Determine the thickness of shim " S_3 ".
3. Split the total shim thickness "Stot" into S_1 and S_2 so that the specified circumferential backlash is present between ring gear and drive pinion.

The aim of this adjustment is to restore the smoothest running position which has been achieved on test equipment in the production line.

To achieve correct results, greatest possible cleanliness for all assembly work and measuring procedures is essential.

When assembling the final drive assembly, it is only necessary to readjust drive pinion and ring gear or drive set if components have been replaced which have a direct influence on the adjustment. Refer to the following table to avoid unnecessary adjustment procedures.

Adjust:	Ring gear ($S_1 + S_2$)	Drive pinion (S_3)
Replaced component		
Transmission case	x	x
Lateral transmission cover	x	
Large cylindrical roller bearing and four-point bearing for drive pinion	x	x
Drive set	x	x
Differential housing	x	
Tapered roller bearing for differential	x	

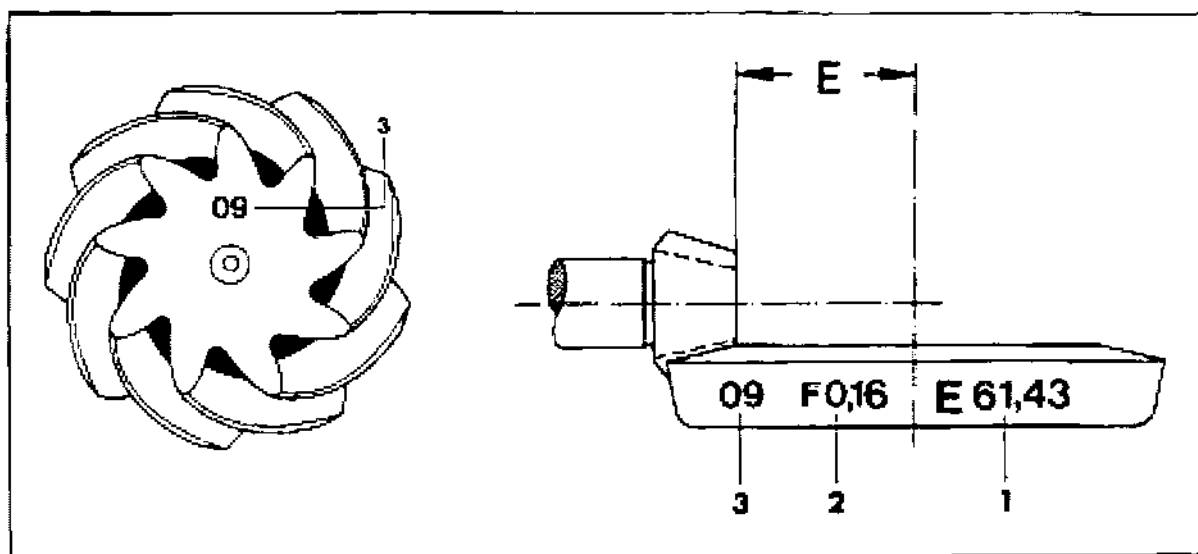


- 1 – Shim S_1
- 2 – Shim S_2
- 3 – Adjusting shim S_3
- E – Setting value

Correct results may only be achieved if assembly work and measuring procedures are carried out carefully and with maximum cleanliness.

General

The setting of drive pinion and ring gear is a determining factor for the service life and smooth running of the rear-axle drive. Drive pinions and ring gears that have been checked for good tooth contact pattern and low noise in both directions of rotation on special test equipment are therefore matched during production. The position at which smoothest running can be achieved is determined by shifting the drive pinion axially, and embossed on the ring gear as setting value "E".



1 = setting "E" (e.g. 61.43 mm)

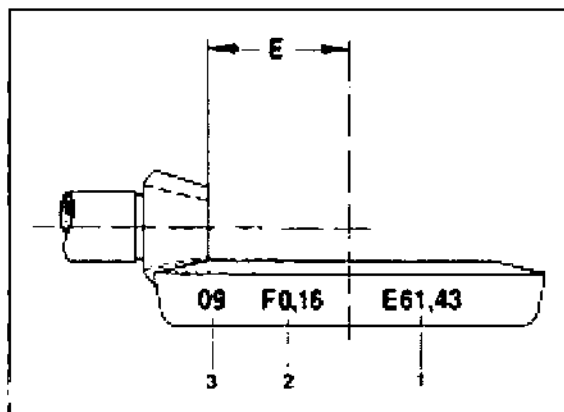
2 = circumferential backlash "F"
(e.g. 0.16 mm)

3 = matching number

39 08 15 Adjusting drive pinion

Note

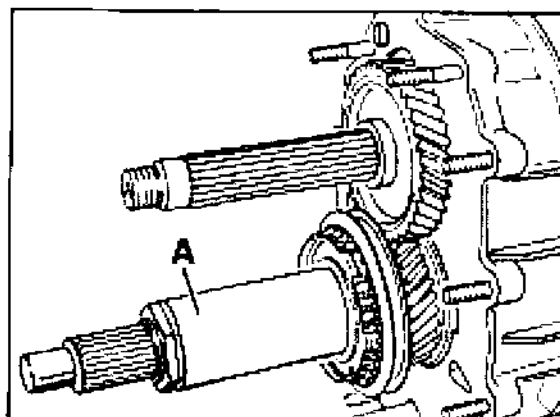
The setting value "E" is indicated on the ring gear.



217-39

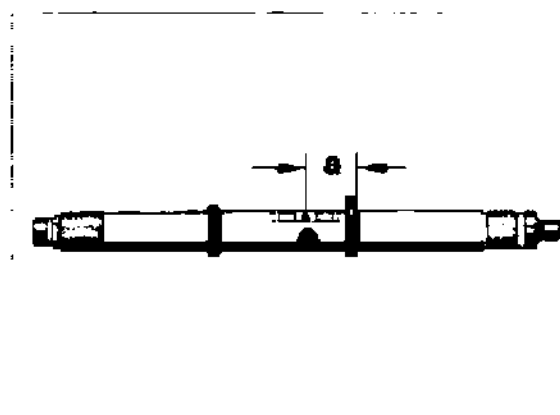
1 = Setting value "E"

1. Install complete gear set without "S3" shims and tighten all tensioning plate hexagon-head nuts to **23 Nm** (17 ftlb.).
2. Fit gear housing and locate with three nuts.
3. Install and engage sixth gear.
4. Block input shaft with Special Tool 9253 and tighten drive pinion collar nut to **300 Nm** (221 ftlb.).



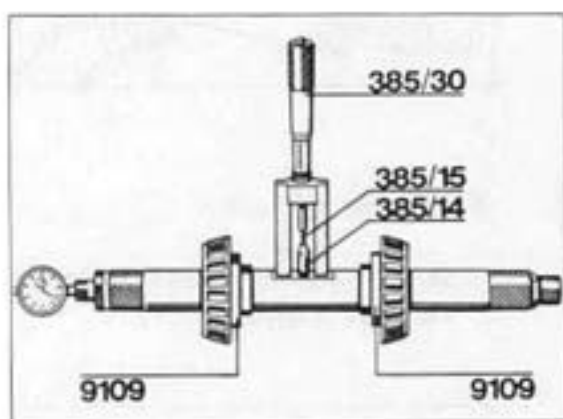
A = Suitable pipe section

5. Set adjusting ring of measuring mandrel **VW 385/1** to dimension "a".



a = approx. 65 mm

6. Assemble measuring mandrel with centering disks **9109**, tapered roller bearing, plunger **VW 385 /14** and dial gauge extension **VW 385/15**.
7. Set measuring mandrel with master gauge **385/30** to the setting value (61.43 mm in the example). Set dial gauge (3 mm measuring range) to zero with 1 mm preload.



543.39

8. Put gauge block plate **9281** on drive pinion head and insert measuring mandrel into transmission case. Dial gauge extension is located in the area of the gauge block plate.



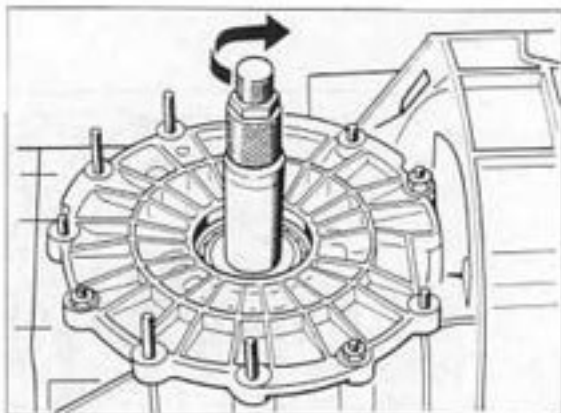
544.39

9. Fit lateral transmission cover without O-ring and tighten crosswise with 4 nuts.

Note

Do not use a hammer when fitting the lateral transmission cover (the gauge block plate held by magnets might fall off). Fit cover in installation position only by tightening the nuts uniformly.

10. Pull second centering disk with spindle towards the outside until the measuring mandrel can just be turned by hand.



545-20

Example:

If the small pointer on the dial gauge is between 1 and 2 and the large pointer indicates 0.37 mm, then 0.37 mm is the shim thickness (with 1mm gauge preload) to be inserted. Always round up or down to the nearest 0.05 mm (e.g. 0.37 mm to 0.35 mm).

12. After inserting the necessary shims, check the setting value "E" again. A deviation of ± 0.03 mm is permissible.

11. Turn measuring mandrel carefully until the dial gauge extension is vertical to the face of the drive pinion head. At this point, the pointer of the dial gauge reaches maximum deflection (reverse point) and the dial gauge must be read.

Note

The measured value always deviates from the set dimension clockwise (the smaller pointer on the dial gauge is between 1 and 2), i.e. if the dial gauge is set with a preload of 1 mm, the value deviating from 1 is taken as shim thickness "S₃".

Adjusting ring gear

Determine total shim thickness "S tot."
($S_1 + S_2$).

The ring gear must be adjusted, if the:

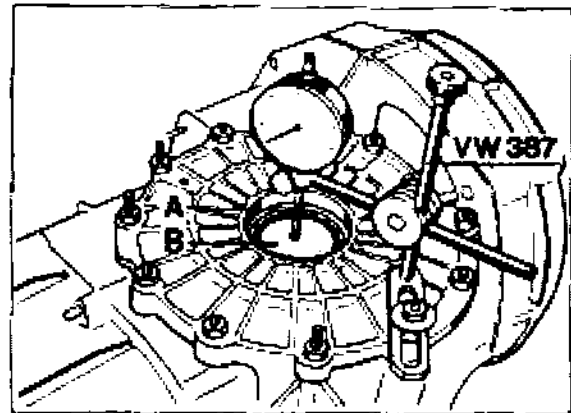
transmission case,
lateral transmission cover,
tapered roller bearing for differential,
differential housing or drive set
have been replaced.

Note

The drive pinion must be removed to determine the preload of the differential tapered roller bearings.

1. Make sure that the bearing outer races of the tapered roller bearings are well seated in the transmission case or lateral transmission cover, respectively.
2. Fit one spacer ring (2.5 mm thick) on the ring gear side and on the opposite side of the differential to be used.
3. Insert differential into transmission case and rotate several times.
4. Fit lateral transmission cover without seal and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).
5. Put gauge block plate **VW 385/17** on the collar of the differential.

6. Fasten universal dial gauge holder **VW 387** with dial gauge and extension to the case and set to 0 with 2 mm preload.



546-39

- A = Dial gauge extension (approx. 30 to 40 mm long)
B = Gauge block plate VW 385/17

7. Move differential up and down. Read off backlash on the dial gauge and note.

Note

Do not turn differential while measuring backlash as this will give an incorrect reading.

8. Calculate "S tot.".

"S tot." = Fitted shim thickness
+ Measured value
+ Pressure fit of tapered roller bearing

Example

Thickness of shims inserted	5.00 mm
Measured value	0.75 mm
Pressing (constant value)	<u>0.40 mm</u>
"S tot."	6.15 mm

9. Remove differential, pull off both tapered roller bearings and split calculated shim thickness "S tot." as follows.

Select spacer S₁ 0.70 mm thinner and S₂ 0.70 mm thicker as a starting point for subsequent adjustment of the backlash.

Example:

Calculated thicknesses

$$S_1 + S_2 = 2.375 + 3.775 = 6.15 \text{ mm}$$

Rounded thicknesses

$$S_1 + S_2 = 2.35 + 3.80 = 6.15 \text{ mm}$$

Measure shims with a micrometer in several places. Permissible deviation 0.02 mm. Also check shims for burrs and damage.

Example

Total shim thickness of spacers

$$S_1 + S_2 = 6.15 \text{ mm}$$

Thickness of spacer S₁

$$\begin{array}{rcl} \frac{6.15 \text{ mm}}{2} & = & 3.075 \text{ mm} \\ & - & 0.700 \text{ mm} \\ \hline & & 2.375 \text{ mm} \end{array}$$

Thickness of spacer S₂

$$\begin{array}{rcl} \frac{6.15 \text{ mm}}{2} & = & 3.075 \text{ mm} \\ & + & 0.700 \text{ mm} \\ \hline & & 3.775 \text{ mm} \end{array}$$

Note

Spacers are available in thicknesses of 1.6 to 3.1 mm in increments of 0.10 mm.

By using a 0.25 mm shim, the shim thicknesses may be graduated in increments of 0.05 mm.

The calculated shim thicknesses must be rounded up or down for plausible dimensions that will not alter the total thickness S₁ and S₂.

Adjusting circumferential backlash

Note

The backlash to be set is embossed on the ring gear.

1. Mount gear set using shims "S₃" determined while adjusting the drive pinion.

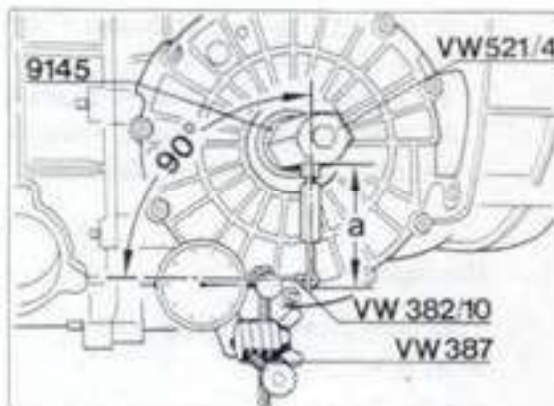
Make sure that the collar nut of the drive pinion is tightened to **300 Nm** (221 ftlb.) before measuring backlash.

2. Insert differential with tapered roller bearing and shims (S₁ + S₂) into the housing.
3. Fit lateral transmission cover and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).

Always make sure that there is a certain amount of backlash when tightening the nuts. Never allow the drive pinion to seize.

4. Assemble measuring lever **VW 388** and adjusting device **VW 521/4** and adjust lever length to 80 mm with the plunger. Refer to dimension "a" in the picture.
5. Insert adjusting device with clamping sleeve (Special Tool **9145**) into the differential and clamp firmly.
6. Rotate differential in both directions several times to settle the tapered roller bearings.

7. Fit universal dial gauge holder with flat extension in such a way as to produce a right angle between dial gauge axis and lever.



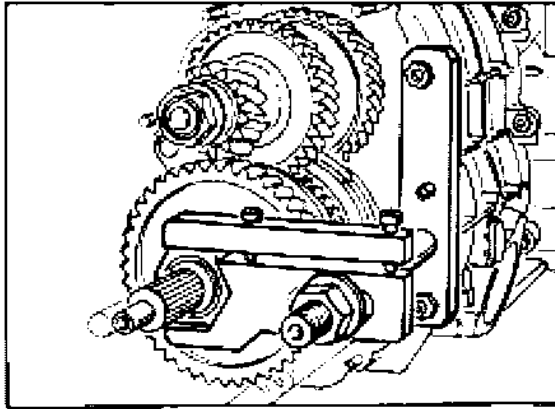
548-29

Dimension "a" = approx. 80 mm

8. Turn ring gear carefully at the clamping screw of the adjusting device as far as the stop and set the dial gauge to zero. Turn back ring gear and read off circumferential backlash. Note down the reading.

Note

When carrying out measurements, the drive pinion must be blocked with Special Tool 9562.



1710-39

9. After turning the ring gear a further 90°, repeat measuring procedures three times.
The measured values must not deviate from one another by more than 0.03 mm.

Note

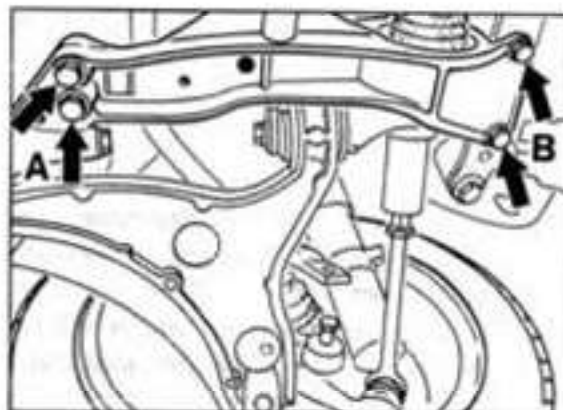
The backlash to be adjusted is embossed on the ring gear. A deviation of ± 0.03 mm is permissible

10. If the required backlash cannot be obtained, replace spacers ($S_1 + S_2$) again.
The total shim thickness ("S tot.") must not be altered, however.

39 88 19 Removing and installing front-axle final drive

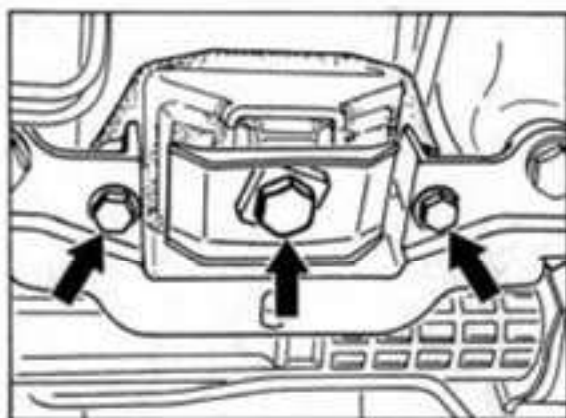
Removal

1. Remove front and center underside panel.
2. Unclip power steering lines from steering gear.
3. Separate drive shafts with Special Tool 9581 at transmission end.
4. Screw out mounting bolts of transmission mount, raise front axle final drive and take out mounts.



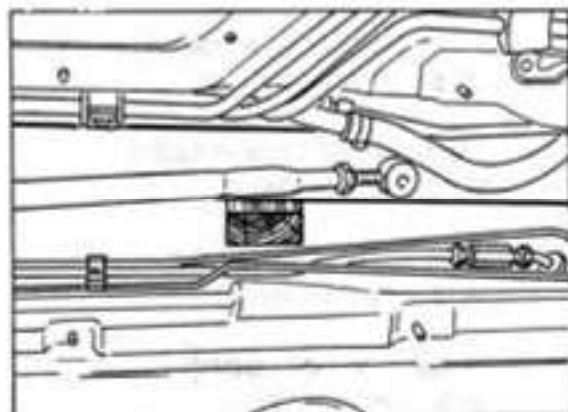
1964-39

- A - Remove bolts
B - Slacken bolts only (approx. 5 mm)



1963-39

5. Lift front-axle final drive and take out final drive mount.
6. Screw out mounting bolts "A" of side member and slacken bolts "B" only (by approx. 5 mm).



1965-39

9. Push front axle final drive out of splines and take out from below at an oblique angle. Take care not to damage the brake pipe at the spare wheel well. Place front axle out of the way so that the pinion shaft faces up.

Note

If body tolerances are unfavorable, slacken mounting bolts of front axle final drive by approx. 5 mm.
This will allow the crossmember and the central tube to move down somewhat and will make removal easier.

Installation

1. Check universal-joint shaft for true running (see page 40 - 103).
2. Check front-axle final drive oil and top up if required.
3. Measure front axle.

4. Tightening torques: (= ftlb)

Front-axle final drive mount to central tube

M 12 85 Nm (63)

Front-axle final drive mount to cross member

M 8 23 Nm (17)

Front-axle final drive to central tube

M 10 46 Nm (34)

Drive shaft to transmission

M 8 42 Nm (31)

Stabilizer to side member

M 8 23 Nm (17)

Stabilizer to stabilizer mount

M 10 46 Nm (34)

Side member to body / cross member
front / rear

M 12 90 Nm (66) / M 10 46 Nm (34)

cross member to body

outer / inner

M 12 105 Nm (77) / M 10 48 Nm (35)

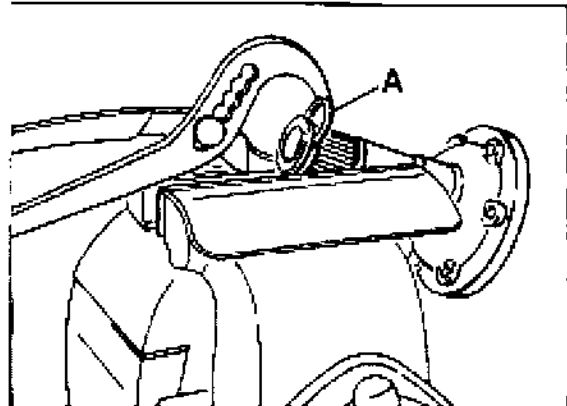
39 59 19 Removing and installing oil seal of halfshaft flange (front axle final drive)

Note

The seals may also be replaced with the front axle final drive remaining installed.

Removal

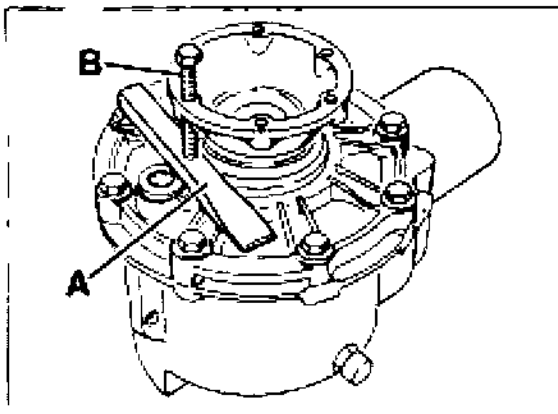
1. Remove halfshaft flange by placing a chisel or spacer „A“ below the halfshaft flange and turning bolt „B“ to pull out halfshaft flange. In case the halfshaft flange is tilted, screw another bolt into the opposite hole.



1994-39

A = New snap ring

3. Grease snap ring liberally and press in halfshaft flange with a suitable drift.



1993-39

2. Lever out seal with a suitable tool.

Installation

1. Pack space between dust and grease lips with grease (e.g. Liqui Moly Pu 53) and drive seal home to stop, using Special Tool 9537.
2. The snap ring of the halfshaft flange must always be replaced. Use protective jaws to mount flange in a vise and press out snap ring with a new snap ring „A“.

39 55 19 Removing and installing output shaft oil seal (Front axle final drive)

Removal

1. Remove front axle final drive (also refer to page 39 - 223).
2. Lever out seal with Special Tool **VW 681**.

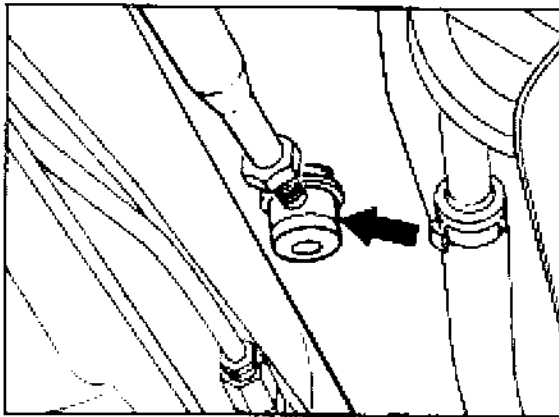
Installation

1. Using Special Tool **9545**, press in seal to stop.

39 03 19 Removing and installing central tube

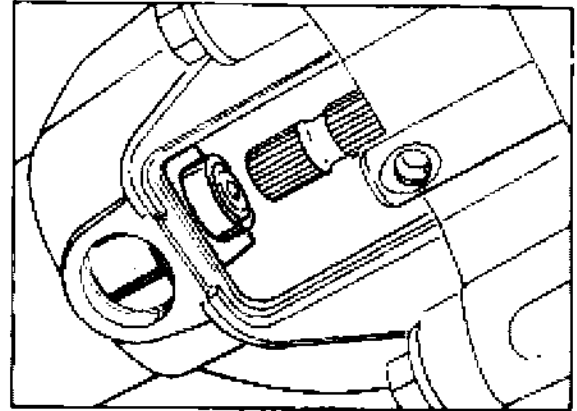
Removal

1. Remove centre console.
2. Remove underside panels.
3. Remove fit bolt for gearshift rod coupling.
4. Lift guide tube off ball joint.

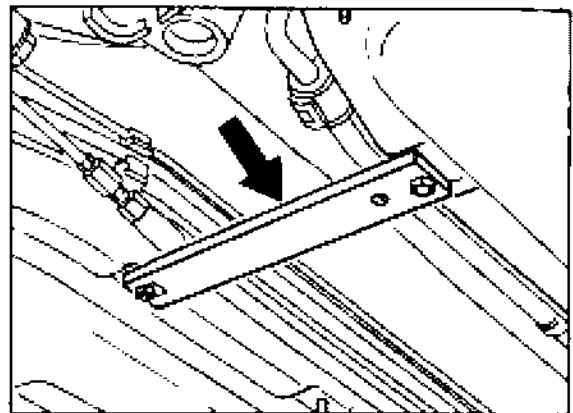


2238-39

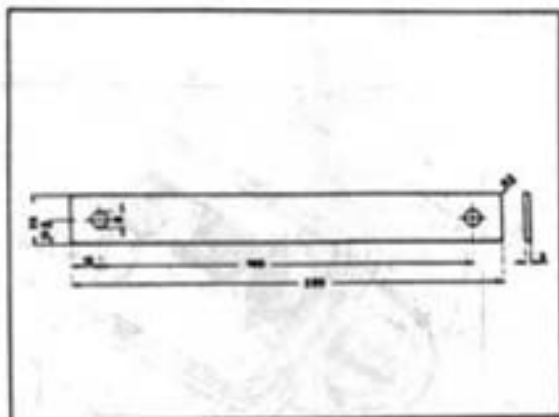
5. Remove shift lever with gearshift rod and guide tube.
6. Remove front axle final drive.
(see page 39 - 225).
7. Completely unscrew the two mounting screws of the clamping sleeve and push it forwards on the central tube.



8. Apply flat steel parts (made in workshop) to support central tube.

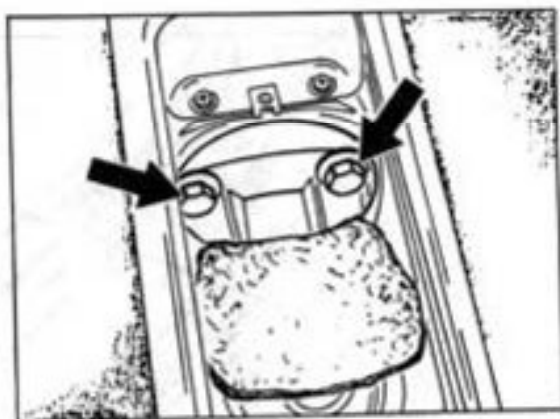


2022-10



2029-10

9. Unscrew manual transmission mounting screws (the top two screws can be reached from the passenger compartment).



2025-10

Two persons are required for the removal of the central tube.

Installation

Torque specifications

Front axle final drive and suspension
(see page 39 - 226)

Frame to tunnel:

M 6 = 10 mm

Gearshift rod to transmission - internal shift rod:

M 8 = 23 Nm (17 ftlb)

Slides to guide tube:

M 5 = 6 Nm (4.5 ftlb)

Double clamp to central tube:

M 8 = 35...40 Nm (26...29.5 ftlb)

Central tube to manual transmission

M 12 = 85 Nm (63 ftlb)

1. The parts removed must be installed in reverse order of removal.
2. Make sure that the centering pins on the transfer casing are properly positioned.



2240-39

3. Place insulation in correct position on tunnel.

4. Apply sliding compound (e.g. Contifix) to central tube in area of insulation and push central tube in carefully, making sure that the insulation is properly positioned.
5. Check adjustment of gearshift.
(see page 34 - 31).